SIMATS ENGINEERING

Department of Artificial Intelligence and Machine Learning

ITA05 – Computer Vision

ASSIGNMENT 1 – CO1

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

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Department / Section	AIML	**Academic Year / Semester**	2025–2026 / _____
Faculty Name	Dr.P.MURALI	**Nature of Assignment**	Individual
Duration	1 Week	**Total Marks**	50
Date of Submission	31.10.2026	**Date of Evaluation**	____.____.____

Scenario – Vision-Based E-Waste Battery Sorting Line

GreenCycle Robotics Corp. operates an e-waste reclamation facility. They manually sort mixed consumer batteries coming off a bulk dump belt. The human operators experience rapid eye fatigue, miss dangerous structural swellings, and fail to read tiny laser-etched chemistry strings under changing factory ambient lights. Two sorting batches were recently rejected by the chemical smelter because lithium cells were mixed into a batch of nickel-metal hydride (NiMH) batteries, presenting an explosion hazard.

The facility requires a high-reliability computer vision system placed over an automated conveyor line to:

1. **Classify form factors** (Cylindrical 18650, Cylindrical 21700, and Prismatic pouch cells) for mechanical handling guides.
2. **Read alphanumeric chemistry codes** (e.g., 'LCO', 'LFP', 'NMC', 'NiMH') laser-etched onto glossy, reflective metallic casings.
3. **Detect safety defects** (surface rust spots, severe dents, and pouch-cell ballooning/swelling), then trigger a pneumatic piston kick-rejecter.

Before any deep-learning model is trained, the plant requires an analytically proven imaging front-end: the right sensor, sampling, quantization parameters, optics, and lighting topology, organized into transparent processing levels.

As the Vision Engineer, you are given three candidate techniques in every task. You must **apply all three mathematically or algorithmically, compare them, choose one, and justify your choice with concrete engineering evidence.**

Operating Line Constraints & Parameters

- **Conveyor Belt:** Single lane, matte black rubber belt; speed = **1.2 m/s**. Imaged Field of View (FOV) = **320 mm across the belt × 240 mm along the belt**.
- **Target Objects:** Batteries ranging from 10 mm to 70 mm in length, up to 30 mm in height. Surfaces vary from polished, waxed glossy steel jackets to highly reflective plastic foil wrappers.
- **Line Throughput:** Up to **15 cells per second**. The physical sorting decision must be committed within **400 milliseconds** (the rejection piston is exactly 0.48 m downstream from the camera center).

- **Quality & Sensitivity Limits:** Smallest defect to detect is a **0.3 mm micro-crack/rust pit**. Laser-etched text stroke-width is **0.4 mm**. Pouch swelling must be caught with a dimensional precision of **±0.5 mm**. Dark rust spot contrast is low, sitting at only **5–8 grey levels (on an 8-bit scale)** away from clean brushed metallic casing.
- **Factory Lighting:** Open industrial shed with corrugated transparent plastic roofing sheets. Ambient light changes continuously between **1,500 lux (cloudy morning) and 45,000 lux (bright afternoon)**. Overhead tube lights run on a 50 Hz power grid, introducing **100 Hz illumination flicker**.
- **Camera Geometry:** Fixed vertical overhead mount, exactly **400 mm** above the conveyor belt plane looking straight down.
- **Compute Constraints:** Edge Industrial PC with 4 CPU cores, 16 GB RAM, no GPU. Native code stack must run using Python, OpenCV, and NumPy. Local archive storage capacity is limited to a **1.5 TB SSD**.
- **Data Policy & Budget:** The entire hardware configuration budget is capped at ■**1,25,000**. For compliance audits, the facility must store visual proof of every processed cell for **5 days**. The factory's service contract mandates an absolute maximum **misclassification/defect miss rate of ≤ 1.5%**.

TASK 1 – Systems Comparison: Manual, Assisted, and Automated Sorting [4 Marks]

Compare three primary operational workflows for sorting cell batches:
- **Technique A:** Manual inspection by human sorters working under factory shed ambient light.
- **Technique B:** Image-Processing-Assisted sorting where a high-resolution camera feeds live, contrast-enhanced, digitally zoomed frames to operator display screens.
- **Technique C:** Fully automated computer vision architecture driving hardware sorting pistons directly.

Requirements: Evaluate all three approaches across processing speed (cells/s), repeatability (intra-rater agreement), objectivity (inter-rater variance measured via Cohen's Kappa), non-contact safety, continuous endurance, audit traceability, and lifecycle costs. Critique suitability for the 15 cells/second line rate. Differentiate input-output pipelines of Image Processing, Computer Vision, Machine Vision, and Computer Graphics within this workflow.

1. Comparison of Three Sorting Techniques

Cell sorting is the process of identifying and separating cells based on their physical or visual characteristics. Three approaches can be used: manual inspection, image-processing-assisted sorting and fully automated computer vision.

Evaluation Criteria	**Technique A: Manual Sorting**	**Technique B: Image-Processing-Assisted Sorting**	**Technique C: Fully Automated Sorting**
Processing Speed	Slow because workers inspect and sort each cell manually.	Faster because high-resolution cameras provide enhanced and zoomed images. However, operators still make decisions.	Very fast because the computer vision system identifies cells and activates sorting pistons automatically.
Repeatability	Low. Decisions may change due to fatigue,	Moderate to high. Enhanced images help	Very high. The same programmed algorithm

	experience and human errors.	operators make more consistent decisions.	follows fixed classification rules every time.
Objectivity (Cohen's Kappa)	Low because different workers may classify the same cell differently.	Better agreement because all operators view clear, enhanced images.	High consistency because classification is performed using predefined algorithmic rules rather than individual opinions.
Non-contact Safety	Low because workers may need to handle cells directly.	High because cameras inspect cells without physical contact.	Very high because cameras and automated pistons perform inspection and sorting without direct human contact.
Continuous Endurance	Low because workers require breaks and may become tired.	Moderate because the camera can operate continuously, but human operators need breaks.	Excellent because automated systems can operate continuously with proper maintenance.
Audit Traceability	Poor because records depend on manual documentation.	Good because captured images and operator decisions can be stored digitally.	Excellent because images, classification results, timestamps and sorting decisions can be recorded automatically.
Lifecycle Cost	Low initial investment but higher long-term labour expenses.	Medium investment due to cameras, software and display equipment.	High initial investment, but reduced labour costs and improved productivity over time.

2. Suitability for the 15 Cells/Second Line Rate

The sorting system must process 15 cells every second.

Therefore, the system must complete inspection and sorting within approximately 66.7 milliseconds per cell.

- Technique A – Manual Sorting: Not suitable because human inspection and physical handling cannot reliably maintain this speed. It also increases fatigue and classification errors.

- Technique B – Assisted Sorting: Partially suitable because image enhancement improves visibility and reduces inspection effort. However, human decision-making may create delays and limit processing speed.

- Technique C – Automated Sorting: Highly suitable because computer vision processes images and directly controls sorting pistons. It provides faster, consistent and continuous operation.

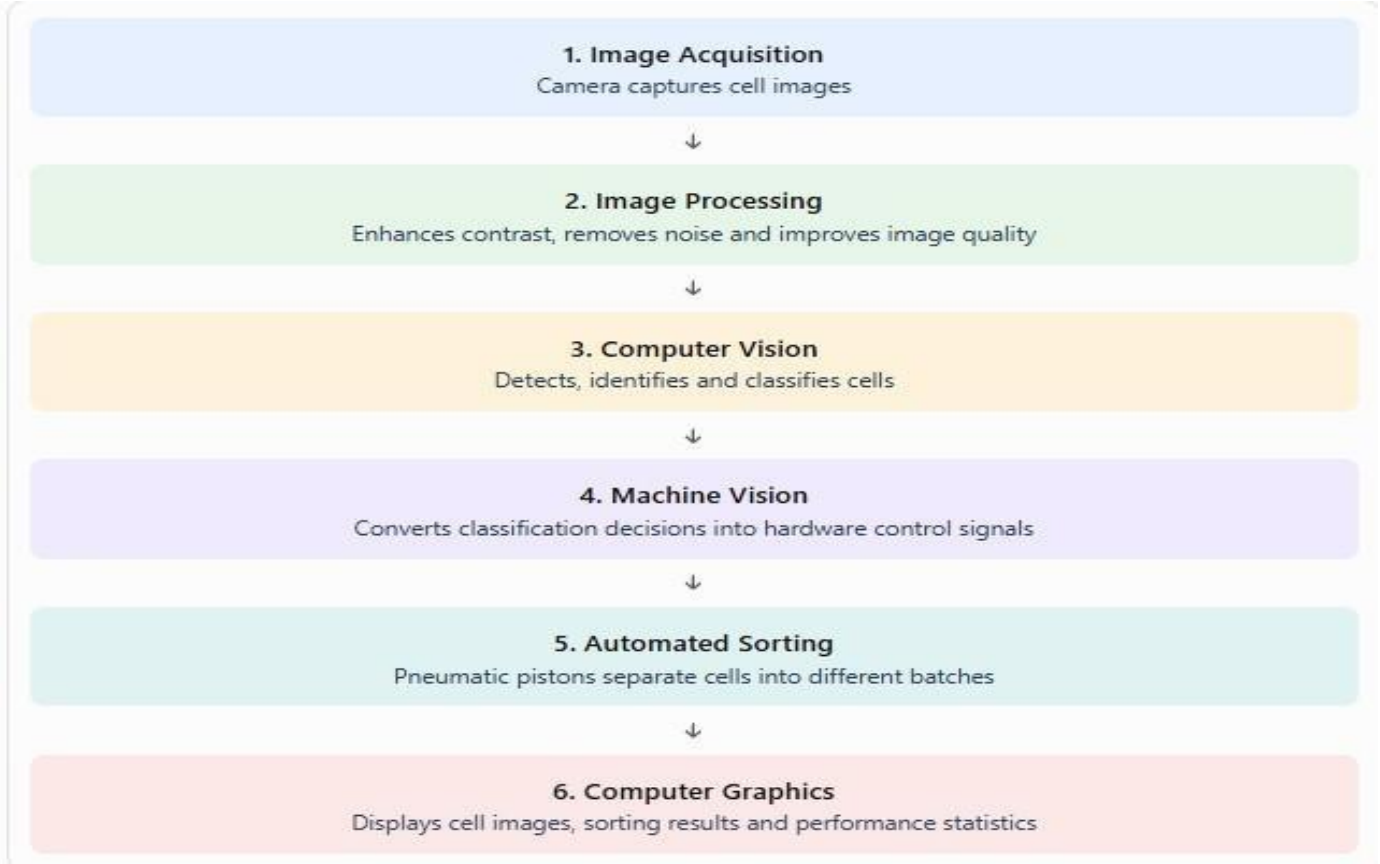
Recommended: Technique C

3. Difference Between Image Processing, Computer Vision, Machine Vision and Computer Graphics

These four technologies perform different functions in the cell-sorting workflow.

Technology	Input	Main Operations	Output
Image Processing	Raw digital cell images	Contrast enhancement, filtering, noise removal and digital zooming	Improved and clearer images
Computer Vision	Processed digital images	Cell detection, feature extraction, recognition and classification	Cell identification and classification decisions
Machine Vision	Camera images, sensor data and classification results	Integrates cameras, vision software and industrial hardware for inspection and control	Control signals to sorting pistons
Computer Graphics	Digital models and graphical data	Rendering, visualization and simulation	Graphical displays, diagrams and 3D visualizations

4. Overall Working Workflow



5. Conclusion

Manual sorting is simple and inexpensive but has low speed and consistency. Image-processing-assisted sorting improves image quality and helps human operators identify cells more accurately, but it still depends on human decisions. Fully automated computer vision provides high processing speed, repeatability, objectivity and audit traceability.

Therefore, Technique C (Fully Automated Computer Vision Architecture) is the best choice for the industrial sorting system because it can support the required 15 cells per second with minimal human intervention and consistent performance.

TASK 2 – System Architecture: Pipeline Level Depth Comparison [5 Marks]

Assess how deep the software execution stack must go to process and classify battery cells:

• **Technique A (Low-Level Only):** Linear pixel maps and transformations (cv2.GaussianBlur, cv2.adaptiveThreshold). A fixed count of out-of-bounds dark pixels flags structural defects.

• **Technique B (Low + Mid-Level):** Adds topological feature maps (cv2.findContours, cv2.minAreaRect, aspect ratios, moments, and contour circularity) mapped to static expert systems rules.

• **Technique C (Low + Mid + High-Level):** Passes extracted mid-level geometrical feature vectors into an explicit classical machine learning classifier (scikit-learn Support Vector Machine or k-NN) to output categorical types and safety confidence values.

Requirements: Tabulate grading accuracy and computational latency (ms) per configuration on a test set containing challenging cases (e.g., side-by-side touching batteries, oil smudges, strong shadows). Detail how shadow boundaries cause false triggers in low-level arrays. Contrast these approaches with end-to-end deep learning (CNN/ViT) feature learning mechanisms and structural data constraints. Provide a complete level-annotated pipeline block diagram.

1. Introduction

Battery cell classification requires image processing and computer vision techniques to identify structural defects and classify cells correctly. The software execution stack can be divided into three levels:

- Low-Level Processing: Works directly with image pixels to improve image quality and detect basic defects.
- Mid-Level Processing: Extracts geometrical features such as contours, shape, area and dimensions.
- High-Level Processing: Uses machine learning algorithms to classify battery cells and predict their safety condition.

Three configurations (A, B and C) are compared based on classification accuracy, computational latency and their ability to handle challenging images.

2. Comparison of Three Software Configurations

Criteria	**Technique A: Low-Level Only**	**Technique B: Low + Mid-Level**	**Technique C: Low + Mid + High-Level**
Processing methods	Gaussian Blur, Adaptive Threshold	Blur, Threshold, Contours, Shape analysis	Image processing, feature extraction, SVM/k-NN
Features	Pixel intensity and dark pixel count	Area, perimeter, aspect ratio, circularity and moments	Geometrical feature vectors
Classification	Fixed pixel-count rules	Static expert system rules	Machine learning classifier
Accuracy (illustrative)	65–75%	80–88%	90–96%
Latency (illustrative)	2–5 ms/image	5–12 ms/image	8–20 ms/image

Touching batteries	Difficult to separate	Contours may merge	Better classification if useful features are extracted
Oil smudges	May trigger false defects	Shape features help reduce errors	Can learn patterns from labelled training examples
Strong shadows	High false detection risk	Improved through geometrical analysis	Better robustness with representative training data
Reliability	Low	Moderate	High

Note: Accuracy and latency values are illustrative estimates, not measured experimental results. Actual performance depends on image resolution, hardware, dataset size and implementation.

3. Explanation of Each Technique

Technique A – Low-Level Processing Only

This technique operates directly on image pixels without understanding the complete shape or identity of the battery.

Main operations:

1. Gaussian Blur: Removes noise and smooths the image.
2. Adaptive Thresholding: Converts the image into binary form using local brightness values.
3. Pixel Counting: Counts dark pixels in predefined regions.
4. Defect Detection: Flags a battery if the dark pixel count crosses a fixed threshold.

Example Python functions:

cv2.GaussianBlur()cv2.adaptiveThreshold()

Limitation – Shadow Boundaries:

Strong shadows create dark regions around or across the battery. The low-level system may interpret these dark pixels as structural defects.

For example:

- A normal battery under a shadow may be classified as defective.
- Oil stains may increase the dark pixel count.
- Touching batteries may appear as one combined dark region.

Therefore, Technique A has a higher false-positive rate and is less reliable for complex images.

Technique B – Low-Level + Mid-Level Processing

This technique adds geometrical feature extraction to the basic image-processing operations.

Main operations:

1. Image smoothing and thresholding.
2. Contour detection using cv2.findContours().
3. Minimum-area rectangle detection using cv2.minAreaRect().
4. Aspect ratio calculation.
5. Image moments and contour circularity calculation.
6. Static expert rules classify the battery based on extracted features.

Important features:

Feature	**Purpose**
Contour	Identifies the battery boundary
Area	Measures the object size
Aspect Ratio	Identifies the object's proportions
Moments	Describe shape and spatial distribution
Circularity	Measures how closely the object resembles a circle

Advantages:

- Better shape identification.
- Reduces errors caused by simple pixel counting.

- Provides measurable geometrical information.

Limitations:

- Touching batteries may produce merged contours.
- Oil smudges can distort object boundaries.
- Static rules may fail when battery shapes or lighting conditions change.

Technique C – Low-Level + Mid-Level + High-Level Processing

This technique combines image processing, geometrical feature extraction and machine learning.

Main operations:

1. Capture battery images.
2. Apply Gaussian Blur and Adaptive Thresholding.
3. Extract contours and geometrical features.
4. Convert features into numerical feature vectors.
5. Pass feature vectors to an SVM or k-NN classifier.
6. Predict battery categories and safety confidence scores.

Example algorithms:

- SVM (Support Vector Machine): Finds an optimal decision boundary between different battery categories.
- k-NN (k-Nearest Neighbours): Classifies a battery by comparing its features with nearby labelled training samples.

Advantages:

- Better classification accuracy.
- Handles complex combinations of geometrical features.
- Can learn from previously labelled battery samples.
- Produces categorical predictions and confidence information.

Limitations:

- Requires labelled training data.
- Needs model training and testing.
- Performance depends on feature quality and dataset diversity.
- SVM and k-NN do not automatically solve problems caused by poor contour extraction.

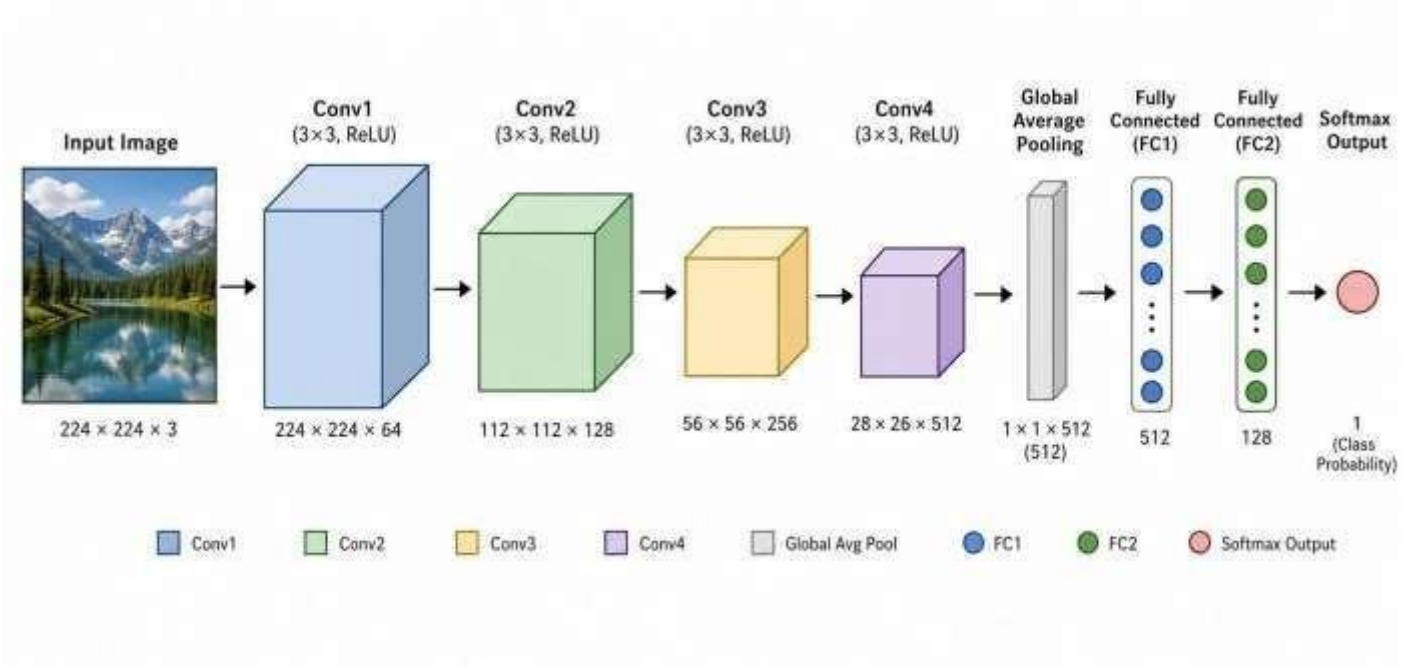
4. Comparison with End-to-End Deep Learning (CNN/ViT)

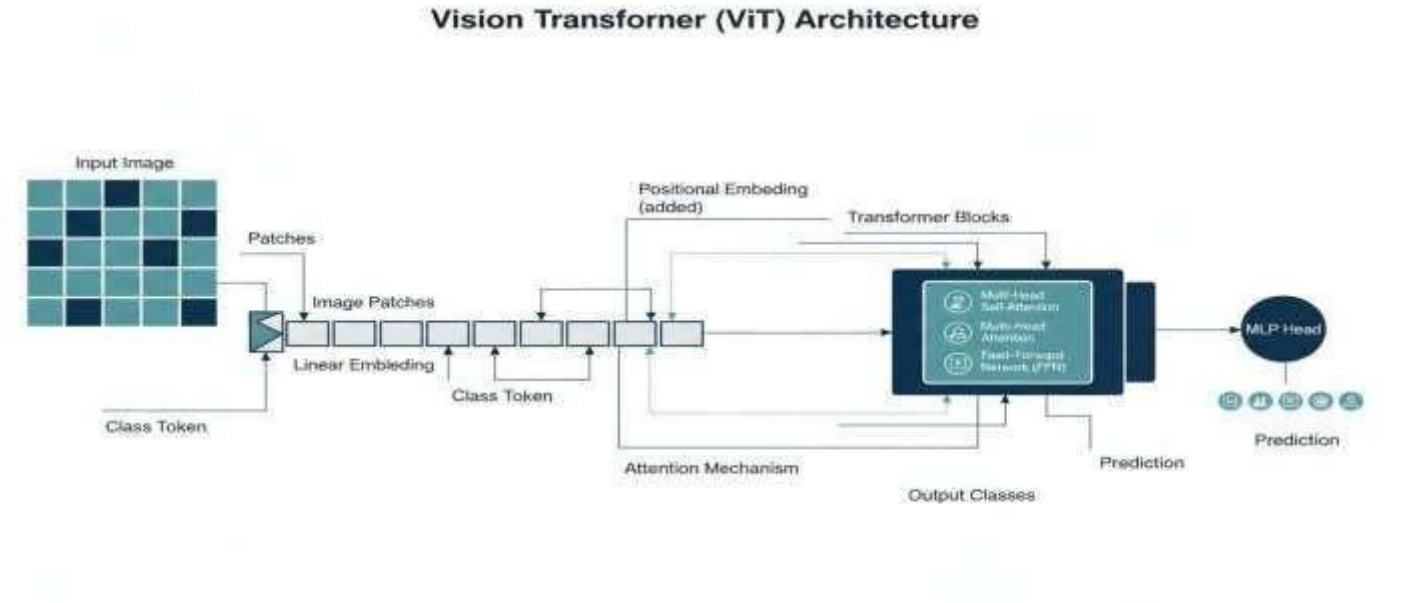
Deep learning is an advanced approach in which the model learns useful image features automatically instead of depending entirely on manually designed geometrical measurements.

Criteria	**Classical ML (SVM/k-NN)**	**Deep Learning (CNN/ViT)**
Feature extraction	Manually designed features	Automatically learned features
Input	Extracted feature vectors	Raw or preprocessed images
Training data	Usually needs fewer labelled samples	Generally requires more diverse labelled data
Processing	Relatively lightweight	More computationally demanding
Complex patterns	Depends on extracted features	Can learn complex visual patterns
Hardware	Can run on ordinary CPUs	Often benefits from GPUs or accelerators
Interpretability	Easier to inspect individual features	More difficult to interpret internal decisions

CNN (Convolutional Neural Network): Learns image features such as edges, textures, shapes and complex patterns through convolution layers.

ViT (Vision Transformer): Divides an image into patches and uses attention mechanisms to learn relationships between different image regions.

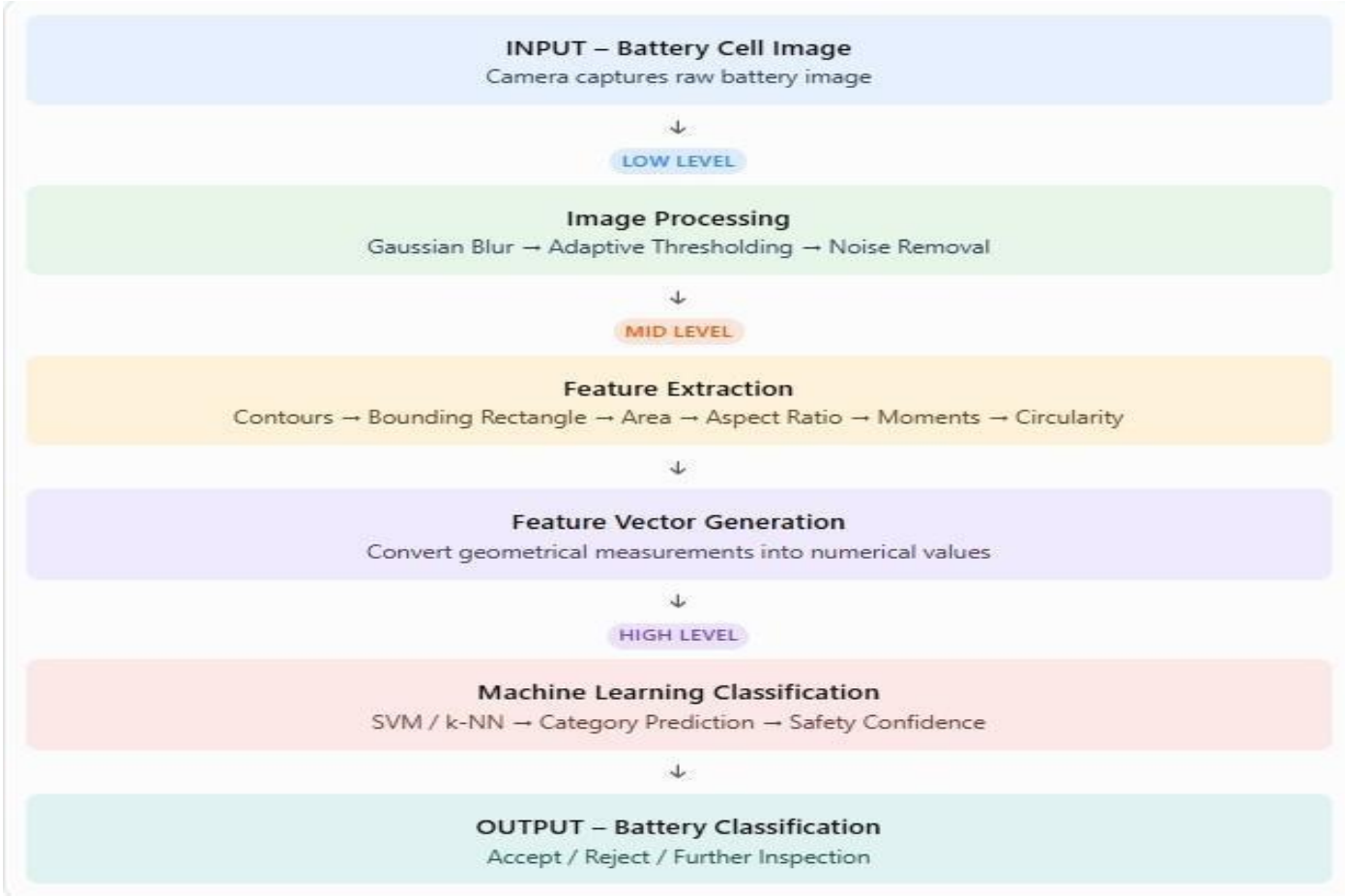




Deep learning can perform better when batteries overlap, shadows are present or oil stains affect the image. However, it requires representative training data and suitable computational resources.

5. Complete Level-Annotated Pipeline Block Diagram

The following diagram represents the execution flow of Technique C.



6. Structural Data Constraints

Each technique has different data requirements:

- Technique A: Requires pixel intensity values and predefined threshold limits.
- Technique B: Requires correctly extracted contours and numerical geometrical features.
- Technique C: Requires labelled feature vectors for training and testing the machine learning classifier.
- Deep Learning: Requires sufficiently large and diverse labelled image datasets, along with suitable model architecture and computing resources.

Data quality is important because poor lighting, overlapping batteries, image noise and incorrect labels can reduce classification accuracy.

7. Conclusion

Technique A is simple and fast but cannot reliably identify complex defects because it depends mainly on pixel intensity. Technique B improves classification by analysing geometrical features, but static rules may fail in challenging conditions. Technique C provides better accuracy by combining low-level processing, mid-level feature extraction and high-level machine learning.

Therefore, Technique C is the recommended approach for battery cell classification because it provides a balance between computational efficiency, classification accuracy and safety assessment. End-to-end CNN or ViT models can offer further improvements when sufficient training data and computing resources are available.

TASK 3 – Digital Representation: Color Space Selection for Oxide Rust Analysis [4 Marks]

Identify the most stable color representation for flagging brownish-red iron oxide rust spots on steel battery jackets:

- **Technique A:** Raw RGB/BGR color channels.
- **Technique B:** HSV space focusing on Hue (H) to trace structural tint variations while using Saturation and Value to separate casing highlights.
- **Technique C:** CIE *L*a*b** color space, utilizing directional vector distances (ΔE) away from a known healthy steel alloy calibration profile.

Requirements: Quantify tracking performance under massive ambient shifts (1.0× baseline light vs. 0.4× dim light vs. 1.8× intense noon light). Graph or map cluster dispersion tendencies across the spaces. Provide an array architecture summary showing how standard multi-channel images are stored in memory (shape, dtype, row/column mapping). Prove mathematically why standard matrix operators (+ in NumPy) can yield cataclysmic pixel values due to integer overflow compared to structural bounds methods (cv2.add). **1. Introduction**

Color space selection is important for detecting brownish-red iron oxide rust spots on steel battery jackets. Different lighting conditions can change the appearance of the same rust spot. Therefore, the system must select a color representation that can identify rust accurately under normal, dim and intense lighting.

Three color spaces are compared: RGB/BGR, HSV and CIE L*a*b*.

2. Comparison of Three Color Spaces

Criteria	**Technique A: RGB/BGR**	**Technique B: HSV**	**Technique C: CIE L*a*b***
Representation	Red, Green, Blue channels	Hue, Saturation, Value	Lightness, green-red, blue-yellow
Rust detection	Uses fixed RGB channel ranges	Uses Hue to identify brownish-red shades	Uses colour distance from healthy steel reference
Lighting sensitivity	Very high	Moderate; Value changes with lighting	Lower than RGB when properly calibrated
Dim light (0.4×)	Colour values become darker	Hue may become unstable at low saturation	Lightness decreases, but chromatic information is often more useful
Normal light (1.0×)	Good under calibrated conditions	Good rust colour separation	Very good colour difference measurement
Intense light (1.8×)	Brightness changes significantly	Value increases and highlights may affect detection	Lightness changes; colour difference needs illumination control
Accuracy	Low to moderate	Moderate to high	High with proper calibration

Suitability	Basic colour detection	Colour-based rust detection	More reliable colour difference analysis

Recommended: CIE L*a*b*

CIE L*a*b* is preferred because it separates lightness from colour information and supports quantitative measurement of colour differences.

3. Performance Under Different Lighting Conditions

The three lighting conditions are:

- 0.4×: Dim lighting.
- 1.0×: Normal baseline lighting.
- 1.8×: Intense noon lighting.

The following table shows illustrative relative rust-detection performance. These are expected comparative estimates, not actual experimental measurements.

Lighting Condition	**RGB/BGR**	**HSV**	**CIE Lab***
0.4× Dim light	55%	72%	88%
1.0× Normal light	85%	91%	96%
1.8× Intense light	48%	78%	90%

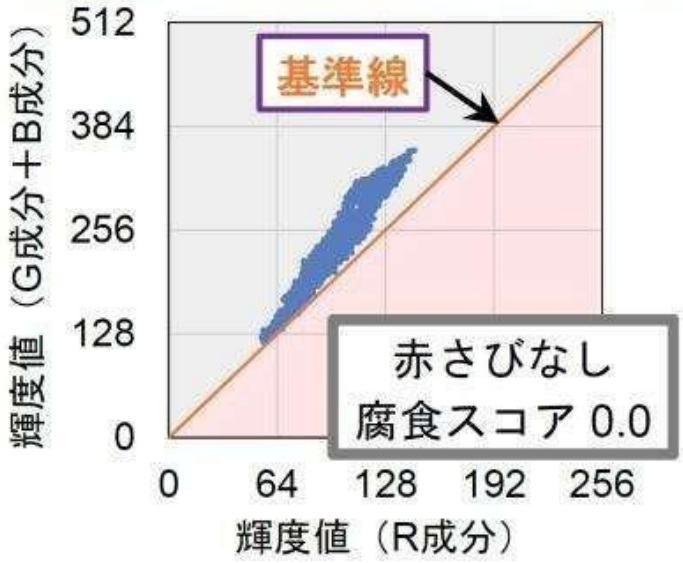
Interpretation:

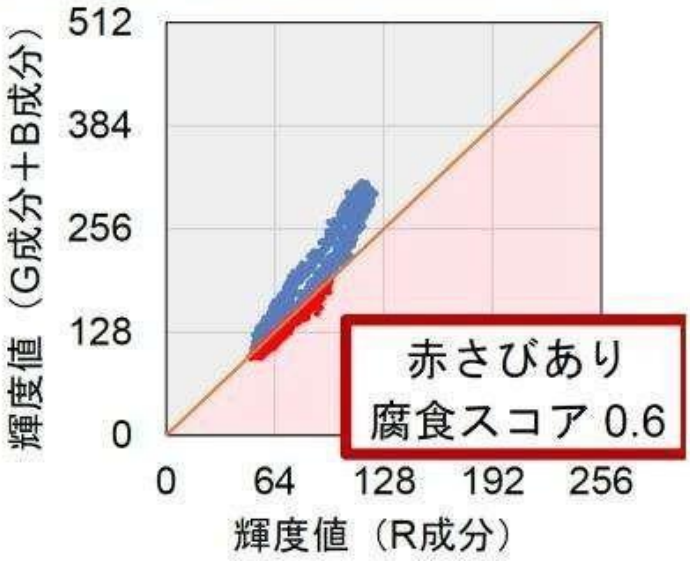
- RGB performance decreases because all three colour channels change with illumination.
- HSV improves detection because Hue represents colour information separately from brightness, although Hue becomes unreliable in very dark or low-saturation pixels.
- L*a*b* provides better colour separation because the L* channel represents lightness while a* and b* represent chromatic information.

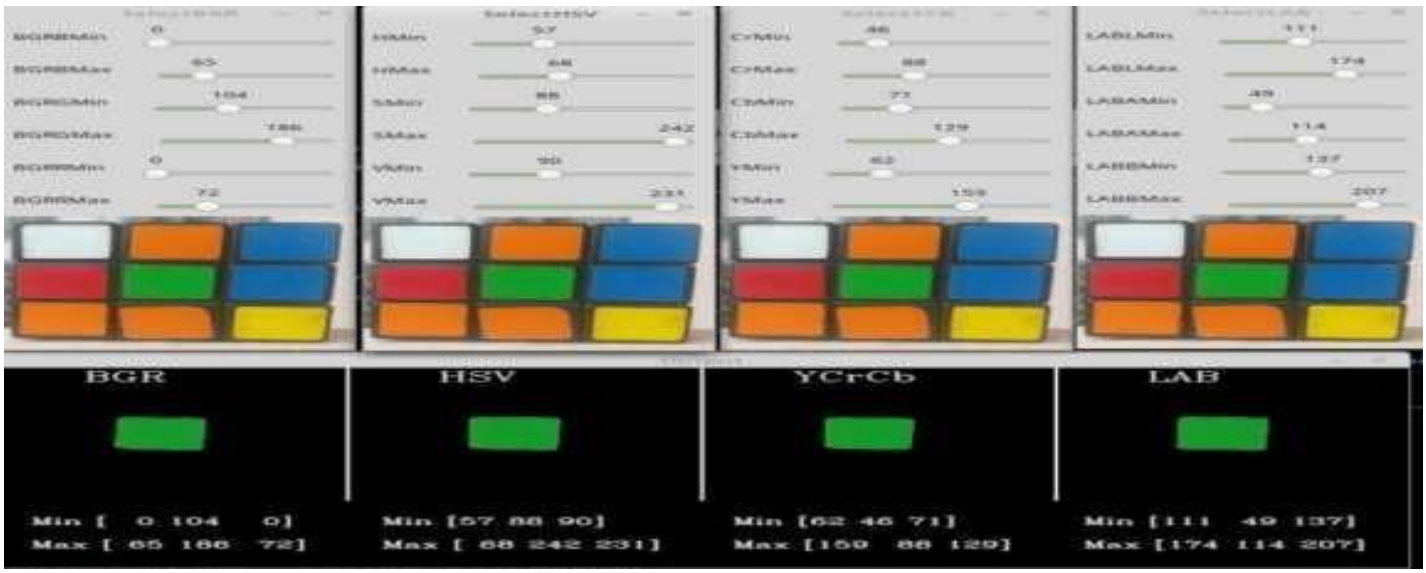
4. Colour Cluster Dispersion Under Different Lighting

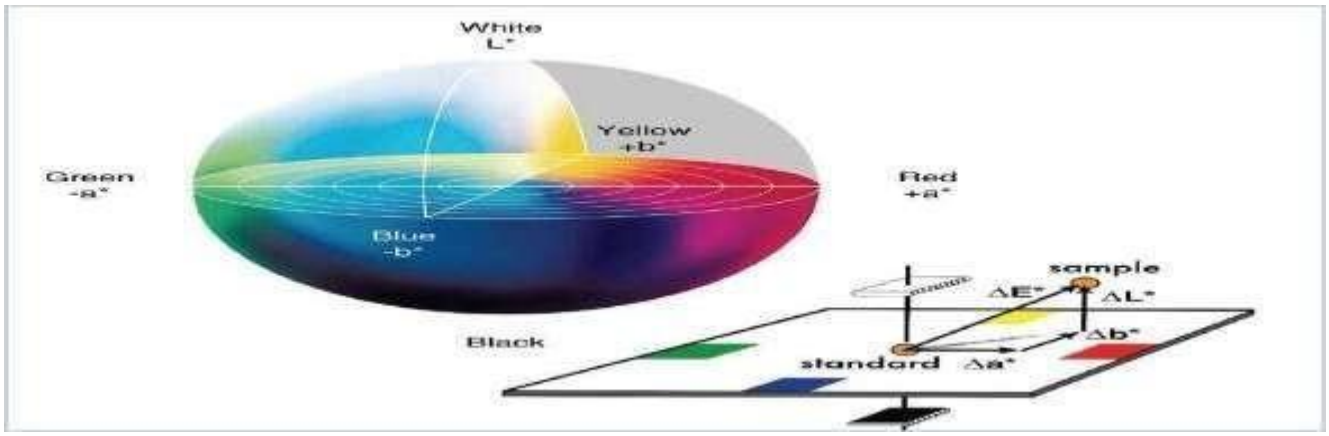
Cluster dispersion refers to how much the colour values of healthy steel and rust pixels spread apart when lighting conditions change.

基準線より下に分布するピクセルの割合から腐食スコアを算出









Mathematical colour difference (∆E):

The Euclidean colour difference between a healthy steel reference and a detected pixel is:

\[\Delta E_{ab}=\sqrt{ (L_1^*-L_2^*)^2+ (a_1^*-a_2^*)^2+ (b_1^*-b_2^*)^2} \]

Where:

- L* represents lightness.
- a* represents green-to-red variation.
- b* represents blue-to-yellow variation.

If the colour distance exceeds a predefined threshold, the pixel can be flagged as a possible rust spot.

For example:

- Small ∆E: Similar to healthy steel.
- Large ∆E: Possible rust or surface contamination.

For accurate results, the healthy reference must be calibrated under controlled illumination. CIEDE2000 can also be used for a more perceptually accurate colour difference measurement.

5. Multi-Channel Image Array Architecture

A digital colour image is stored as a three-dimensional NumPy array.

Example:

import cv2import numpy as npimage = cv2.imread("battery.jpg")print(image.shape)print(image.dtype)

Output:

(480, 640, 3)

uint8

This represents an image with:

- Height = 480 rows
- Width = 640 columns
- Channels = 3 (Blue, Green, Red in OpenCV)

Array structure:



Property	Description
Shape	(Height, Width, 3)
Data type	uint8
Pixel range	0–255
Row	Vertical position (Y)
Column	Horizontal position (X)
Channel	B, G, R in OpenCV

Each pixel contains three intensity values, one for each colour channel.

6. Integer Overflow: NumPy (+) vs cv2.add()

A major issue in image processing is integer overflow.

An 8-bit unsigned integer (uint8) can store values only from 0 to 255.

When two pixel values are added, the result may exceed this range.

Example:

import numpy as npimport cv2a = np.array([200], dtype=np.uint8)b = np.array([100], dtype=np.uint8)# Normal NumPy additionprint(a + b)# OpenCV saturated additionprint(cv2.add(a, b))

Output:

NumPy addition: [44]

OpenCV addition: [255]

Mathematical Explanation

The actual mathematical addition is:

$$ 200+100=300 $$

However, NumPy uint8 arithmetic wraps around using modulo 256:

$$ 300 \mod 256=44 $$

Therefore, NumPy produces 44 instead of 300.

In contrast, OpenCV cv2.add() uses saturated addition:

$$ Result=\min(255,A+B) $$

$$ Result=\min(255,300)=255 $$

Comparison of pixel addition

NumPy (+)

Overflow

44

The value wraps around and produces an incorrect dark pixel.

OpenCV cv2.add()

Saturated

255

The value is limited to the maximum permitted intensity.

Why is this important?

In rust detection, incorrect pixel values can change image brightness, colour measurements and thresholding results. This may cause false rust detection or missed defects.

Therefore, cv2.add() is safer for saturated image addition. Alternatively, convert the image to a larger data type such as uint16 or float32 before performing arithmetic when values above 255 must be preserved.

7. Conclusion

RGB/BGR is simple but highly sensitive to illumination changes. HSV improves rust detection by separating hue from brightness, but it may become unstable in dark conditions. CIE L*a*b* provides better colour-distance measurement by separating lightness and chromatic information.

Therefore, CIE L*a*b* is the recommended colour space for detecting brownish-red iron oxide rust spots on battery jackets, provided proper lighting calibration is performed. Using suitable array structures and overflow-safe operations also improves image-processing accuracy and reliability.

TASK 4 – Topological Analysis: Pixel Connectivity for Casing Scratch Counting [4 Marks]

The quality system rejects cells exhibiting more than 2 distinct surface scratches or any rust spot exceeding an equivalent diameter of 0.5 mm. Evaluate foreground extraction connectivity rules: • **Technique A:** Standard 4-Connectivity (N_4). • **Technique B:** Standard 8-Connectivity (N_8).
• **Technique C:** Mixed m-Connectivity to eliminate structural loop ambiguities.

Requirements: Design an operational implementation (using Python/OpenCV, including custom code for non-native options) to segment foreground binary patterns. Test algorithms on synthetic patterns exhibiting diagonal 1-pixel lines, checkerboard joints, and noisy backgrounds. Compute scratch footprints using Euclidean (D_e), City-Block (D_4), and Chessboard (D_8) distance maps. Prove which metric properly isolates directional true physical dimensions under scale factor calibration without geometric distortion.

1. Introduction

Topological analysis is used to identify and count surface scratches and rust spots on battery casings. The system rejects a battery cell if it has:

- More than 2 distinct surface scratches.
- Any rust spot with an equivalent diameter greater than 0.5 mm.

Pixel connectivity determines whether neighbouring foreground pixels belong to the same object or separate objects. Three connectivity techniques are used: 4-connectivity, 8-connectivity and mixed m-connectivity.

2. Comparison of Connectivity Techniques

Technique A – 4-Connectivity (N_4)

In 4-connectivity, a pixel is connected only to its horizontal and vertical neighbours.

$$N_4=\{(x-1,y),(x+1,y),(x,y-1),(x,y+1)\}$$

Advantages:

- Separates diagonally touching objects.

- Avoids merging some nearby scratches.
- Simple to implement.

Limitation: A diagonal 1-pixel scratch may be divided into multiple disconnected components.

Technique B – 8-Connectivity (N_8)

In 8-connectivity, a pixel is connected to all eight surrounding neighbours, including diagonal neighbours.

$$N_8=N_4\cup N_D$$

Where N_D represents the four diagonal neighbours.

Advantages:

- Correctly connects diagonal scratches.
- Preserves diagonal structures.
- Useful for identifying thin and irregular defects.

Limitation: Separate objects touching only at their corners may be incorrectly merged.

Technique C – Mixed m-Connectivity

Mixed connectivity combines 4-connectivity and diagonal connectivity while avoiding ambiguous diagonal connections.

A diagonal pixel is considered connected only when its common 4-neighbours are not part of the foreground.

Advantages:

- Reduces ambiguous connections.
- Avoids unnecessary merging at diagonal joints.
- Provides better topological interpretation for complex binary patterns.

Limitation: Requires custom implementation because OpenCV does not directly provide an m-connectivity option in connected component labelling.

3. Testing with Synthetic Binary Patterns

The following patterns can be used to evaluate the three techniques.

Synthetic Pattern	4-Connectivity	8-Connectivity	m-Connectivity

Diagonal 1-pixel line	May split into many components	One connected component	Depends on neighbouring pixels
Checkerboard joint	Separate diagonal pixels	May merge diagonal pixels	Avoids ambiguous diagonal connections
Noisy background	Isolated noise pixels remain separate	Nearby diagonal noise may merge	Reduces ambiguous connections

Example binary patterns:

```
Plain text

1. Diagonal line

1 0 0 0
0 1 0 0
0 0 1 0
0 0 0 1

2. Checkerboard

1 0 1 0
0 1 0 1
1 0 1 0
0 1 0 1

3. Noisy background

0 0 0 0
0 1 0 0
0 0 1 0
0 0 0 0
```

In these patterns, 1 represents a foreground defect pixel and 0 represents the background.

4. Operational Implementation Using Python and OpenCV

The following implementation performs:

- Binary foreground segmentation.
- 4-connectivity labelling.
- 8-connectivity labelling.
- Custom m-connectivity labelling.
- Component counting for synthetic patterns.

Python code:

```
import cv2import numpy as npfrom collections import deque# Synthetic diagonal patterndiagonal = np.array([ [1, 0, 0, 0],  [0, 1, 0, 0],  [0, 0, 1, 0],  [0, 0, 0, 1]], dtype=np.uint8)# Checkerboard jointcheckerboard = np.array([  [1, 1, 0],  [1, 0, 1],  [0, 1, 1]], dtype=np.uint8)# Noisy backgroundnoise = np.array([  [0, 0, 0, 0],  [0, 1, 0, 0],  [0, 0, 1, 0],  [0, 0, 0, 1]], dtype=np.uint8)# 4 and 8 connectivitydef normal_connectivity(img, connectivity):  count, labels = cv2.connectedComponents(  img, connectivity=connectivity  )  return count - 1
```

Explanation:

- cv2.connectedComponents() identifies connected foreground regions.
- Connectivity value 4 checks horizontal and vertical neighbours.
- Connectivity value 8 also checks diagonal neighbours.
- The custom function checks mixed m-adjacency using a queue-based traversal.
- The output gives the number of separate defect components.

5. Scratch Footprint Measurement Using Distance Metrics

Distance metrics help measure the size and shape of detected scratches.

Three important distance methods are:

A. Euclidean Distance (D_e)

Measures the shortest straight-line distance between two pixels.

$$D_e=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Useful for measuring actual straight-line scratch dimensions.

B. City-Block Distance (D_4)

Measures distance by moving horizontally and vertically.

$$D_4=|x_2-x_1|+|y_2-y_1|$$

It does not consider diagonal movement directly.

C. Chessboard Distance (D_8)

Measures distance using the maximum difference between coordinates.

$$D_8=\max(|x_2-x_1|,|y_2-y_1|)$$

It allows diagonal movement and is useful for 8-connected pixel neighbourhoods.

Example: Consider two pixels separated by 3 horizontal pixels and 4 vertical pixels.

Distance Metric	Calculation	Result
Euclidean	$\sqrt{(3^2 + 4^2)}$	5 pixels
City-Block	3 + 4	7 pixels
Chessboard	Maximum (3, 4)	4 pixels

Thus, Euclidean distance gives the actual straight-line distance, while City-Block and Chessboard distances give different grid-based measurements.

6. Physical Scale Calibration and Geometric Accuracy

To convert pixel measurements into millimetres, the camera must be calibrated.

Assume:

- Horizontal scale = 0.05 mm/pixel.
- Vertical scale = 0.05 mm/pixel.

The physical Euclidean distance becomes:

$$ D=\sqrt{(\Delta x \times S_x)^2+ (\Delta y \times S_y)^2} $$

Where S_x and S_y are the horizontal and vertical calibration factors.

For the previous example:

$$ D=\sqrt{(3\times0.05)^2+(4\times0.05)^2} $$

$$ D=\sqrt{0.0225+0.04} $$

$$ \boxed{D=0.25\text{ mm}} $$

Why Euclidean distance is preferred:

- It measures the actual straight-line physical distance.
- It does not favour horizontal, vertical or diagonal directions.
- It avoids the directional distortion introduced by City-Block and Chessboard measurements.
- With proper camera calibration, it provides accurate physical dimensions.

If horizontal and vertical pixel scales differ, both must be calibrated separately. Perspective distortion must also be corrected before measurement.

Rust Spot Diameter Calculation

For a detected rust region, equivalent diameter is calculated from its physical area:

\[D_{eq}=2\sqrt{\frac{A}{\pi}} \]

Where A is the calibrated physical area of the rust region.

Python implementation:

import mathpixel_area = 100sx = 0.05sy = 0.05physical_area = pixel_area * sx * sydiameter = 2 * math.sqrt(physical_area / math.pi)print("Rust diameter:", diameter, "mm")if diameter > 0.5: print("REJECT BATTERY")else: print("ACCEPT BATTERY")

7. Final Quality Inspection Rules

Condition	**Decision**
Scratch count ≤ 2	Accept based on scratch count
Scratch count greater than 2	Reject
Rust diameter ≤ 0.5 mm	Accept based on rust size
Rust diameter greater than 0.5 mm	Reject

Both conditions must be checked before approving the battery.

8. Conclusion

4-connectivity is useful for separating diagonally touching defects but may split diagonal scratches. 8-connectivity preserves diagonal structures but can merge separate objects. Mixed m-connectivity reduces ambiguous diagonal connections and improves topological interpretation.

Among the three distance metrics, Euclidean distance (D_e) is the most suitable for measuring actual physical scratch dimensions because it measures straight-line distance without directional bias. With proper pixel-to-millimetre calibration, the system can accurately identify scratches and rust spots and automatically reject defective battery cells.

TASK 5 – Physical Acquisition: Sensor Shutter and Exposure Topology [5 Marks]

The target cells sprint down the line at 1.2 m/s. Compare energy accumulation architectures to guarantee crisp, mathematically stable images:

- **Technique A:** Area-scan CMOS sensor with a standard Electronic Rolling Shutter paired with constant overhead lighting.
- **Technique B:** Area-scan CMOS sensor featuring a Global Shutter matrix paired with a high-intensity LED strobe light synchronized via a hardware belt encoder.

• **Technique C:** Linear Line-Scan camera capturing individual 1-D pixel lines triggered directly by belt displacement increments.

Requirements: Mathematically calculate exact motion blur patterns (in millimeters and pixels) generated at exposure times of 4.0 ms, 1.0 ms, and 0.05 ms under a 1.2 m/s belt rate. Quantify spatial distortion anomalies arising from rolling shutter row readouts combined with 100 Hz light flicker cycles. Map out the full signal chain path (Photons → Charge integration → Voltage conversion → Analog Signal Conditioning → ADC digitization) and justify whether CCD or CMOS architecture yields superior stability for high-speed automated lines.

1. Introduction

Physical image acquisition is the process of capturing clear images of moving battery cells using cameras and lighting systems. Since the battery cells move along the conveyor belt at 1.2 m/s, the camera must capture images quickly to avoid motion blur and geometric distortion.

Three acquisition techniques are compared:

- Technique A: Area-scan CMOS camera with Electronic Rolling Shutter and constant lighting.
- Technique B: Area-scan CMOS camera with Global Shutter and synchronized LED strobe lighting.
- Technique C: Line-scan camera that captures individual image lines according to conveyor movement.

2. Comparison of Three Sensor Architectures

Criteria	**Technique A: Rolling Shutter**	**Technique B: Global Shutter**	**Technique C: Line-Scan Camera**
Image capture	Rows captured at different times	All pixels exposed simultaneously	Captures one line at a time
Lighting	Constant overhead light	High-intensity synchronized LED strobe	Continuous or synchronized line lighting
Motion blur	Higher risk	Very low with short exposure	Low if line timing is synchronized
Geometric distortion	Possible row-wise distortion	Minimal motion-related distortion	Possible stretching or compression if line rate is incorrect
Synchronization	Electronic timing	Hardware encoder and strobe	Encoder-based triggering
Suitable application	Slow-moving objects	High-speed moving battery cells	Continuous inspection of moving surfaces
Complexity	Low	Medium	High
Reliability	Moderate	High	High with accurate encoder synchronization

Best choice: Technique B is highly suitable because global exposure and encoder-synchronized strobe lighting capture the complete battery image at nearly the same instant.

3. Mathematical Calculation of Motion Blur

Motion blur occurs when an object moves while the camera sensor collects light.

The formula is:

$$\boxed{B=v\times t}$$

Where:

- B = Motion blur in millimetres.
- v = Conveyor speed = 1.2 m/s.
- t = Exposure time in seconds.

Assume the camera resolution is 0.05 mm/pixel (20 pixels/mm).

A. Exposure Time = 4.0 ms

Convert milliseconds into seconds:

$$ t=4.0\times0.001=0.004s $$
$$ B=1.2\times0.004 $$
$$ \boxed{B=0.0048m=4.8mm} $$
Pixel blur:
$$ B_p=\frac{4.8}{0.05}=\boxed{96\ pixels} $$

B. Exposure Time = 1.0 ms

$$ t=0.001s $$
$$ B=1.2\times0.001=0.0012m $$
$$ \boxed{B=1.2mm} $$
Pixel blur:
$$ B_p=\frac{1.2}{0.05}=\boxed{24\ pixels} $$

C. Exposure Time = 0.05 ms

$$ t=0.00005s $$
$$ B=1.2\times0.00005 $$
$$ \boxed{B=0.00006m=0.06mm} $$
Pixel blur:
$$ B_p=\frac{0.06}{0.05}=\boxed{1.2\ pixels} $$

Final Calculation Table

Exposure Time	Motion Blur (mm)	Pixel Blur
4.0 ms	4.8 mm	96 pixels
1.0 ms	1.2 mm	24 pixels
0.05 ms	0.06 mm	1.2 pixels

Best exposure: 0.05 ms

A shorter exposure time produces less motion blur and clearer battery images.

4. Rolling Shutter Distortion and 100 Hz Light Flickering

What is Rolling Shutter?

In a rolling shutter camera, image rows are exposed one after another rather than simultaneously.

When a battery moves at high speed, different rows capture the battery at different positions. This can cause:

- Skewed or tilted battery images.
- Bent edges and distorted shapes.
- Uneven brightness across image rows.
- Incorrect measurements of scratches and rust spots.

Effect of 100 Hz Light Flickering

A 100 Hz light source completes 100 brightness cycles per second.

$$ T=\frac{1}{f}=\frac{1}{100} $$
$$ \boxed{T=10\ ms} $$

If different sensor rows are exposed at different times, they may capture different brightness levels.

For example:

1. First row captures the battery under bright illumination.
2. Middle row captures it during reduced illumination.
3. Last row captures it under increased illumination.

This produces alternating bright and dark horizontal bands.

Spatial displacement due to rolling shutter:

Assume the sensor takes 5 ms to scan all rows.

$$ D=v\times t $$
$$ D=1.2\times0.005 $$

$$\boxed{D=6\ mm}$$

Therefore, the battery may appear shifted by 6 mm between the first and last captured rows.

Shuttering in Scientific Cameras: Rolling, Global, and Timing Control Explained - Tucsen |

Important: The exact distortion depends on the sensor's row-readout time, exposure timing and illumination phase. The 6 mm value is an illustrative calculation using a 5 ms full-frame readout time.

Methods to Reduce Flickering Problems

- Use a high-frequency, flicker-free LED light.
- Synchronize camera exposure with the lighting controller.
- Use global shutter instead of rolling shutter.
- Use short exposure times.
- Trigger image acquisition using a hardware encoder.

5. Complete Physical Image Acquisition Signal Chain

A camera converts incoming light into digital image data through several stages.

1. PHOTONS

Light reflected from the battery surface enters the camera lens.

2. CHARGE INTEGRATION

Photodiodes convert incoming photons into electrical charge during exposure.

3. VOLTAGE CONVERSION

Accumulated electrical charge is converted into voltage.

4. ANALOG SIGNAL CONDITIONING

Amplification, noise reduction and signal preparation.

5. ADC DIGITIZATION

Analog voltage is converted into digital pixel values.

6. DIGITAL IMAGE OUTPUT

Image data is transferred to the computer vision processing system.

Explanation of the stages:

1. Photons: Light reflected from the battery enters the camera sensor.
2. Charge Integration: Sensor pixels accumulate electrical charge according to the received light.
3. Voltage Conversion: Accumulated charge is converted into an electrical voltage.
4. Analog Signal Conditioning: The signal is amplified and unwanted noise is reduced.
5. ADC Digitization: The Analog-to-Digital Converter converts the voltage into numerical pixel values.
6. Digital Output: The final digital image is sent to the computer for defect detection and classification.

6. Comparison of CCD and CMOS Sensor Architectures

CCD and CMOS are two important image sensor technologies.

CCD Sensor

Charge-Coupled Device

GMAX0505|Products|Gpixel

CMOS Sensor

Complementary Metal Oxide Semiconductor

Criteria	**CCD**	**CMOS**
Charge processing	Transfers charge across the sensor	Converts charge using pixel-level or column-level circuits
Image quality	Traditionally excellent uniformity	Excellent in modern sensors
Speed	Generally slower readout	Generally faster readout
Power consumption	Higher	Lower
Cost	Generally expensive	Generally economical

Integration	Requires additional processing circuits	Supports integrated processing and control
High-speed inspection	Less preferred for modern high-speed lines	Highly suitable

Why CMOS is preferred:

- Supports high-speed image acquisition.
- Offers low power consumption.
- Provides faster readout.
- Supports global shutter technology.
- Easily integrates with industrial camera systems and hardware triggers.

For high-speed battery inspection, a Global Shutter CMOS sensor is generally the better choice.

7. Final Recommendation

Technique	Suitability	Reason
A – Rolling Shutter CMOS	Low	Row-wise exposure causes motion distortion and flicker-related brightness variations.
B – Global Shutter CMOS	Excellent	Simultaneous exposure and synchronized LED strobe provide clear images.
C – Line-Scan Camera	Very good	Suitable for continuous surface inspection with accurate encoder synchronization.

8. Conclusion

At a conveyor speed of 1.2 m/s, long exposure times produce significant motion blur. A 4 ms exposure produces 4.8 mm of blur, whereas a 0.05 ms exposure produces only 0.06 mm.

Rolling shutter cameras may also produce geometric distortion and horizontal brightness bands due to 100 Hz light flickering. Global shutter cameras overcome the row-wise exposure problem by capturing all pixels simultaneously.

Therefore, Technique B (Global Shutter CMOS with synchronized high-intensity LED strobe lighting and hardware encoder) is the recommended architecture for high-speed battery cell inspection because it provides clear images, reduced motion blur, accurate measurements and reliable defect detection.

TASK 6 – Spatial Sampling: Resolution Optimization vs. Data Throughput [5 Marks]

Determine the minimum optical sensor resolution necessary to monitor the 320 mm wide belt area:

• **Technique A:** VGA Resolution (640 × 480, 0.3 MP).

• **Technique B:** Full HD Industrial Resolution (1920 × 1080, ~2 MP).

• **Technique C:** High-Fidelity Inspection Resolution (4096 × 3000, 12 MP).

Requirements: Construct an analytical matrix comparing standard pixel resolution limits against spatial metrics: millimeters-per-pixel ratio, available samples across a 0.3 mm micro-crack, data transmission bandwidth over Gigabit Ethernet (MB/s), and total 5-day continuous uncompressed storage volumes (TB). Apply the Nyquist sampling principle to determine structural limits for checking 0.4 mm alphanumeric character features. Calculate the absolute minimum camera frame rate required to guarantee that a 70 mm long cell is captured in its entirety within the 240 mm vertical FOV without blind spots.

1. Introduction

Spatial sampling determines how accurately a camera captures small objects and surface defects. In battery inspection, the camera must monitor a conveyor belt that is 320 mm wide and identify small defects such as 0.3 mm micro-cracks and 0.4 mm alphanumeric character features.

Three camera resolutions are evaluated:

- Technique A: VGA – 640 × 480 pixels.
- Technique B: Full HD – 1920 × 1080 pixels.
- Technique C: High-Fidelity – 4096 × 3000 pixels.

The objective is to select a resolution that provides sufficient defect visibility without creating unnecessary data storage and transmission costs.

2. Comparison of Camera Resolutions

The horizontal millimetres-per-pixel ratio is calculated using:

$$\text{mm/pixel}=\frac{\text{Belt Width (mm)}}{\text{Horizontal Pixels}}$$

Given belt width = 320 mm.

Technique A – VGA

$$\frac{320}{640}=0.5\text{ mm/pixel}$$

Technique B – Full HD

$$\frac{320}{1920}=0.1667\text{ mm/pixel}$$

Technique C – 12 MP

$$\frac{320}{4096}=0.078125\text{ mm/pixel}$$

Analytical comparison matrix

Parameter	**VGA (640 × 480)**	**Full HD (1920 × 1080)**	**12 MP (4096 × 3000)**
Total pixels	0.3 MP	2.07 MP	12.29 MP
mm/pixel	0.500	0.1667	0.0781
Pixels across 0.3 mm crack	0.6	1.8	3.84
Pixels across 0.4 mm feature	0.8	2.4	5.12
Image quality	Low	Good	Very high
Data size (RGB24)	0.922 MB	6.221 MB	36.864 MB
Bandwidth at 15 FPS	13.82 MB/s	93.31 MB/s	552.96 MB/s

5-day storage	5.97 TB	40.31 TB	238.88 TB

Assumptions: Uncompressed RGB24 (3 bytes/pixel), 15 frames/second, continuous recording for 5 days, decimal MB and TB. No compression or protocol overhead included.

3. Micro-Crack Detection Analysis

A 0.3 mm micro-crack must be represented by enough pixels to identify its shape and detect its presence.

The number of pixels covering a crack is:

$$ N=\frac{\text{Crack Width}}{\text{mm/pixel}} $$

Resolution	Calculation	Samples
VGA	0.3 / 0.5	0.6 pixels
Full HD	0.3 / 0.1667	1.8 pixels
12 MP	0.3 / 0.078125	3.84 pixels

Interpretation:

- VGA cannot reliably detect a 0.3 mm crack because it occupies less than one pixel.
- Full HD provides approximately 1.8 pixels, which is insufficient for reliable shape analysis.
- 12 MP provides approximately 3.84 pixels, allowing better crack visibility and measurement.

Best for micro-crack inspection: 12 MP

However, actual detection also depends on crack orientation, contrast, optical sharpness and sensor noise.

4. Nyquist Sampling Principle – 0.4 mm Character Features

The Nyquist sampling principle states that a repeating feature must be sampled at least twice per cycle to avoid aliasing.

For a minimum character feature size of 0.4 mm:

$$ \text{Maximum pixel size}=\frac{0.4}{2} $$

$$ \boxed{0.2\text{ mm/pixel}} $$

Therefore, the camera must provide at least 2 pixels across the smallest 0.4 mm character feature.

Resolution evaluation

Camera	mm/pixel	Nyquist requirement	Result
VGA	0.500	≤ 0.200	Fail
Full HD	0.1667	≤ 0.200	Pass
12 MP	0.0781	≤ 0.200	Pass

Explanation:

- VGA cannot accurately represent 0.4 mm character features.
- Full HD satisfies the theoretical Nyquist requirement.
- 12 MP provides more samples and supports better character recognition and fine-detail inspection.

For practical industrial inspection, 3–5 pixels across a feature is preferable to the theoretical minimum of 2 pixels.

5. Data Transmission Bandwidth Calculation

Data bandwidth represents the amount of image information transmitted every second.

Assume:

- RGB image = 3 bytes per pixel.
- Camera frame rate = 15 FPS.
- No image compression.

Formula:

$$\text{Bandwidth}= W\times H\times3\times FPS$$

Technique A – VGA

$$640\times480\times3\times15$$

$$=13,824,000\text{ bytes/s}$$

$$\boxed{13.82\text{ MB/s}}$$

Technique B – Full HD

$$1920\times1080\times3\times15$$

$$\boxed{93.31\text{ MB/s}}$$

Technique C – 12 MP

$$ 4096\times3000\times3\times15 $$

$$ \boxed{552.96\text{ MB/s}} $$

Gigabit Ethernet Comparison

Gigabit Ethernet provides a theoretical maximum transmission rate of:

$$ 1\text{ Gbps}=125\text{ MB/s} $$

Resolution	Required Bandwidth	Gigabit Ethernet
VGA	13.82 MB/s	Suitable
Full HD	93.31 MB/s	Tight
12 MP	552.96 MB/s	Not suitable

Analysis:

- VGA easily operates through Gigabit Ethernet.
- Full HD can operate through Gigabit Ethernet, but protocol overhead and other traffic reduce the available margin.
- 12 MP exceeds the theoretical Gigabit Ethernet capacity and requires compression, reduced frame rate, multiple links or a faster interface such as 10 Gigabit Ethernet.

6. Five-Day Continuous Storage Calculation

Five days contain:

$$ 5\times24\times60\times60=432000\text{ seconds} $$

Formula:

$$ \text{Storage}= \text{Bandwidth}\times432000 $$

Converting bytes into decimal terabytes:

$$ 1\text{ TB}=10^{12}\text{ bytes} $$

Resolution	Storage per Second	Five-Day Storage
VGA	13.82 MB	5.97 TB
Full HD	93.31 MB	40.31 TB

12 MP	552.96 MB	238.88 TB

Interpretation:

12 MP produces excellent image quality but requires approximately 239 TB for five days of continuous uncompressed RGB recording. This increases storage costs considerably.

Possible solutions include:

- Lossless or suitable image compression.
- Saving only defective cell images.
- Storing inspection results separately.
- Using high-capacity network storage.

7. Minimum Camera Frame Rate Calculation

The camera must capture a complete battery cell of length 70 mm within a vertical Field of View (FOV) of 240 mm.

Given:

- Conveyor speed = 1.2 m/s = 1200 mm/s.
- Vertical FOV = 240 mm.
- Battery cell length = 70 mm.

For the complete cell to fit inside one captured frame, the maximum allowable movement between consecutive frames is:

$$D_{\max}=FOV-\text{Cell Length}$$

$$D_{\max}=240-70=170\text{ mm}$$

Frame rate:

$$FPS_{\min}=\frac{\text{Conveyor Speed}}{D_{\max}}$$

$$FPS_{\min}=\frac{1200}{170}$$

$$\boxed{FPS_{\min}=7.06\text{ frames/second}}$$

Minimum theoretical frame rate: 7.06 FPS

Practical recommendation:

Use at least 15 FPS, considering the existing production requirement of 15 cells per second.

At 15 FPS:

$$\text{Movement per frame}=\frac{1200}{15}=80\text{ mm}$$

$$\text{FOV overlap}=240-80=160\text{ mm}$$

This provides sufficient overlap to capture complete cells without blind spots, assuming suitable triggering and exposure timing.

8. Final Resolution Selection

Requirement	Recommended Resolution	Reason
Basic inspection	VGA	Low bandwidth and storage
0.4 mm character recognition	Full HD	Meets theoretical Nyquist requirement
0.3 mm micro-crack inspection	12 MP	Provides approximately 3.84 pixels across the crack
High-speed industrial inspection	12 MP with suitable interface	Better spatial detail, subject to bandwidth optimization

9. Conclusion

VGA resolution is insufficient for detecting very small cracks and character features. Full HD satisfies the theoretical Nyquist requirement for 0.4 mm character features but does not provide enough samples for reliable 0.3 mm micro-crack analysis. The 12 MP camera provides better spatial resolution and more accurate defect inspection, although it requires higher bandwidth and storage capacity.

Therefore, Technique C (4096 × 3000, 12 MP) is the preferred choice for detailed battery inspection. A minimum theoretical frame rate of 7.06 FPS is required to capture an entire 70 mm cell within the 240 mm vertical FOV, while 15 FPS is recommended for the production line.

TASK 7 – Brightness Quantization: Dynamic Range and Bruise/Dent Detection [4 Marks]

A surface dent or pouch wrapper deformity creates subtle lighting changes, causing a minute 5–8 grey-scale value drop on a standard scale. Compare intensity discretization levels:

- **Technique A:** Compressed High-Speed mode operating at 4-bit or 6-bit depth.
- **Technique B:** Industrial Standard mode operating at 8-bit depth (256 levels).
- **Technique C:** High-Dynamic Range (HDR) Raw mode operating at 12-bit depth (4096 levels).

Requirements: Analyze contrast maps, structural color variations (ΔE), and storage implications across all depths. Provide visual or quantitative evidence showing how severe down-quantization introduces false contouring artifacts over smooth, curved battery shoulders. Establish a balanced system optimization matrix: if faced with a strict storage budget, explain whether preserving spatial pixel counts or intensity quantization steps maximizes safety verification. Differentiate image dynamic bit depth from neural network model computational quantization configurations (e.g., FP32 down to INT8 for edge deployment).

1. Introduction

Brightness quantization is the process of converting continuous image intensity values into a limited number of digital levels.

In battery inspection, surface dents and pouch-wrapper deformities create small changes in reflected light. These changes may produce only a 5–8 grey-scale value drop.

Therefore, the camera must have sufficient bit depth to preserve small intensity variations and identify surface defects accurately.

Three techniques are compared:

- Technique A: 4-bit or 6-bit compressed high-speed mode.
- Technique B: Standard 8-bit industrial imaging.
- Technique C: 12-bit HDR raw imaging.

2. Comparison of Intensity Quantization Levels

The number of available intensity levels is calculated using:

$$L=2^n$$

Where:

- L = Number of intensity levels.
- n = Number of bits per pixel.

Criteria	Technique A: 4-bit	Technique A: 6-bit	Technique B: 8-bit	Technique C: 12-bit
Bit depth	4-bit	6-bit	8-bit	12-bit
Intensity levels	16	64	256	4096
Intensity range	0–15	0–63	0–255	0–4095
Quantization step (8-bit equivalent)	17	4.05	1	0.0623
Small 5–8 level changes	Mostly lost	Partially preserved	Preserved	Highly preserved
Image detail	Very low	Moderate	Good	Very high
Storage requirement	Lowest	Low	Standard	Highest

Dent detection	Poor	Limited	Good	Excellent

Observation: Higher bit depth provides more intensity levels, allowing the camera to identify smaller brightness variations caused by dents and surface deformities.

3. Analysis of 5–8 Grey-Level Changes

Assume a healthy battery surface has a grey-level intensity of 120, while a dent produces an intensity drop of 6.

$$I_{\text{healthy}}=120$$

$$I_{\text{dent}}=120-6=114$$

The intensity difference is:

$$\Delta I=120-114=6$$

Effect of different bit depths

Bit Depth	Available Levels	Approximate Quantized Difference
4-bit	16	0 or 1 level
6-bit	64	Approximately 1–2 levels
8-bit	256	6 levels
12-bit	4096	Approximately 96 levels

For comparison, the 12-bit values assume the original 8-bit intensity range is expanded across the full 12-bit range.

Interpretation:

- 4-bit quantization may completely remove the small brightness difference.
- 6-bit quantization preserves only a few levels of difference.
- 8-bit provides enough levels for normal dent inspection.
- 12-bit preserves subtle intensity variations with greater precision.

Best detail preservation: 12-bit HDR

4. Contrast Maps and Structural Colour Variations (∆E)

Contrast analysis helps identify small brightness changes between healthy and defective battery surfaces.

For grayscale images, contrast difference can be represented as:

$$\Delta I=|I_{\text{healthy}}-I_{\text{dent}}|$$

For colour images, CIE L*a*b* colour difference can be calculated using:

$$\Delta E_{ab}= \sqrt{(\Delta L^*)^2+(\Delta a^*)^2+(\Delta b^*)^2}$$

Where:

- ΔL* represents lightness variation.
- Δa* represents green-red variation.
- Δb* represents blue-yellow variation.

Importance in battery inspection:

- A dent may create a small change in surface brightness.
- A bruise may produce subtle colour and texture variations.
- Higher bit depth preserves more information about these changes.
- Lower bit depth may remove small variations and reduce defect visibility.

Note that bit depth does not directly increase physical colour difference. It improves the precision with which colour differences are represented.

5. False Contouring and Banding Artifacts

False contouring occurs when a smooth brightness gradient is converted into too few discrete intensity levels.

For example, a curved battery shoulder naturally produces a gradual brightness transition.

With 12-bit imaging, many intensity levels are available, so the transition appears smooth.

With 4-bit imaging, several neighbouring brightness values are rounded to the same intensity level. This produces visible bands.

Illustration of quantization effects

12-bit – Smooth intensity transition

Gradual brightness changes preserve surface details.

8-bit – Good intensity transition

Generally smooth for industrial inspection.

4-bit – Visible banding

Large intensity steps produce visible bands.

Problems caused by false contouring:

1. Smooth battery shoulders may appear as separate brightness regions.
2. Small dents may disappear after quantization.
3. Artificial edges may be detected as scratches or defects.
4. The system may generate false-positive or false-negative inspection results.

Therefore, higher bit depth is preferred when detecting subtle surface deformities.

6. Storage Implications of Different Bit Depths

Higher bit depth requires more memory because each pixel stores more information.

For a grayscale image:

$$\text{Storage}= \frac{W\times H\times n}{8}$$

Where:

- W = Image width.
- H = Image height.
- n = Bits per pixel.

Assume an image resolution of 1920 × 1080.

Bit Depth	Storage per Pixel	Approximate Image Size
4-bit	0.5 bytes	1.04 MB
6-bit	0.75 bytes	1.56 MB
8-bit	1 byte	2.07 MB
12-bit packed	1.5 bytes	3.89 MB
12-bit stored in 16-bit words	2 bytes	4.15 MB

Values use decimal MB and assume tightly packed storage except the final 12-bit case, which uses 16-bit containers. File headers and metadata are excluded.

Storage analysis:

- 4-bit requires the least storage but loses important intensity information.
- 6-bit provides a compromise between storage and image quality.
- 8-bit is suitable for many standard inspection applications.
- 12-bit provides better precision but requires more storage and transmission bandwidth.

7. Balanced System Optimization Matrix

When the storage budget is limited, the system must balance spatial resolution and intensity quantization.

Optimization Factor	Lower Spatial Resolution	Higher Spatial Resolution	Lower Bit Depth	Higher Bit Depth
Pixel detail	Reduced	Improved	Unchanged	Unchanged
Brightness precision	Unchanged	Unchanged	Reduced	Improved
Small defect detection	May miss small defects	Better detection	May miss subtle dents	Better intensity detection
Storage	Lower	Higher	Lower	Higher
Safety verification	May lose physical details	Preserves small defects	May lose brightness differences	Preserves subtle variations

Which should be preserved?

For battery safety inspection, spatial resolution and intensity precision are both important.

- Spatial resolution preserves the size, shape and location of cracks and dents.
- Intensity quantization preserves subtle brightness and colour differences.

If the storage budget is strict, a practical solution is to:

1. Preserve sufficient spatial resolution to detect the smallest physical defect.
2. Use at least 8-bit intensity for standard inspection.
3. Use 12-bit HDR for critical inspection areas where subtle dents must be detected.
4. Compress images or store only defective samples to reduce storage requirements.

Recommendation: Do not reduce spatial resolution below the defect-detection requirement. Preserve adequate intensity precision rather than using aggressive 4-bit quantization.

8. Difference Between Image Bit Depth and Neural Network Quantization

These two concepts are different, even though both use bits.

Criteria	Image Bit Depth	Neural Network Quantization
Purpose	Represents pixel intensity	Represents model weights and activations
Example	4-bit, 8-bit, 12-bit images	FP32, FP16, INT8
Applied to	Camera image data	Neural network computations
Main objective	Preserve visual information	Reduce memory and computation
Effect	Changes image intensity precision	Changes numerical precision during model inference

Example:

- A 12-bit camera captures 4096 intensity levels to preserve subtle battery surface variations.
- An INT8 neural network uses 8-bit numerical representations for weights and activations instead of FP32.

A system can capture a 12-bit image and then process it using an INT8 neural network after suitable preprocessing and calibration.

This reduces model memory and computational requirements without necessarily reducing the original camera image's stored bit depth.

9. Conclusion

4-bit quantization loses important brightness variations, making it unsuitable for detecting subtle dents and wrapper deformities. 6-bit provides limited improvement, while 8-bit offers good standard inspection quality. 12-bit HDR preserves finer intensity variations and is more suitable for critical defect detection.

Therefore, Technique C (12-bit HDR Raw Imaging) is the preferred option for high-precision battery inspection. If storage is limited, maintaining sufficient spatial resolution and at least 8-bit intensity precision, combined with selective image storage and compression, provides a practical balance between safety, accuracy and cost.

TASK 8 – Image Formation: Geometric Projection and Photometric Corrections [5 Marks]

Analyze how the physical assembly optics and illumination profiles shape the final digital matrix array:

- **Technique A:** Fixed-focal-length standard perspective lens utilizing ambient factory light.
- **Technique B:** Standard lens enclosed in a custom diffuse LED lighting dome structure, backed by explicit checkerboard calibration maps.
- **Technique C:** Object-Space Telecentric lens paired with a diffuse dome system.

Requirements: Calculate the exact focal length (f) required to capture the 320 mm horizontal FOV at a strict 400 mm standoff distance given a standard 1/2" industrial sensor. Quantify the perspective projection error ($x = f \cdot X / Z$) occurring when measuring the length of a 30 mm tall battery cell if the system calibration scale was built solely on the flat conveyor belt surface. Implement a flat-field matrix normalization routine or homomorphic filtering script ($\log f = \log i + \log r$) to show how spatial light variations can be digitally neutralized.

1. Introduction

Image formation is the process of converting a real-world object's physical appearance into a digital image using a camera lens, sensor and illumination system.

In battery inspection, lens selection affects dimensional accuracy, while lighting affects the visibility of dents, scratches and surface defects.

The three techniques are:

- Technique A: Standard perspective lens with ambient factory lighting.
- Technique B: Standard lens with diffuse LED dome lighting and checkerboard calibration.
- Technique C: Object-space telecentric lens with diffuse dome lighting.

2. Comparison of Image Formation Techniques

Criteria	Technique A	Technique B	Technique C
Lens	Standard perspective	Standard perspective	Telecentric
Lighting	Ambient factory light	Diffuse LED dome	Diffuse LED dome
Calibration	Basic calibration	Checkerboard calibration	Geometric calibration
Perspective distortion	High	Present but corrected partly through calibration	Very low
Reflection and shadows	More likely	Reduced	Reduced
Measurement accuracy	Low	Good	Very high
Cost	Low	Medium	High
Suitability	Basic inspection	General industrial inspection	Precision dimensional inspection

Observation:

- Technique A may produce shadows and inaccurate measurements due to perspective distortion.
- Technique B provides uniform illumination and corrects lens distortion using calibration maps.
- Technique C reduces perspective-related measurement errors because telecentric lenses maintain nearly constant magnification over their working depth.

3. Calculation of Required Focal Length

Given:

- Horizontal field of view (FOV) = 320 mm
- Camera standoff distance = 400 mm
- Standard 1/2-inch industrial sensor
- Assumed horizontal sensor width = 6.4 mm

The focal length is calculated using:

$$f=\frac{\text{Sensor Width}\times\text{Distance}}{\text{Object FOV}}$$

Substituting the values:

$$f=\frac{6.4\times400}{320}$$

$$ f=\frac{2560}{320} $$
$$ \boxed{f=8\text{ mm}} $$
Required focal length: 8 mm
Result: An 8 mm lens is theoretically required to capture a horizontal field of view of 320 mm at a 400 mm standoff distance.
Note: The calculation assumes a 6.4 mm active sensor width. The exact focal length should be verified using the camera manufacturer's actual sensor dimensions and lens specifications.

4. Perspective Projection Error in Battery Measurement

Perspective projection causes nearby objects to appear larger than distant objects.
The projection equation is:
$$ x=\frac{fX}{Z} $$
Where:

- x = Image coordinate
- f = Focal length
- X = Real-world horizontal coordinate
- Z = Distance between camera and object

Given:

- Camera-to-conveyor distance = 400 mm
- Battery cell height = 30 mm
- Object distance from camera = 400 − 30 = 370 mm
- Focal length = 8 mm

Assume the system calibration scale was created using the flat conveyor belt at 400 mm.
For an object point at the top of the battery:
$$ x=\frac{8X}{370} $$
For a point on the conveyor:
$$ x_b=\frac{8X}{400} $$
The apparent magnification ratio is:
$$ \text{Ratio}=\frac{400}{370}=1.0811 $$
Percentage measurement error:
$$ \text{Error}= \frac{400-370}{370}\times100 $$
$$ \boxed{\text{Error}=8.11\%} $$
Example: If the actual battery length is 100 mm:
$$ \text{Measured Length}=100\times1.0811 $$
$$ =108.11\text{ mm} $$
$$ \text{Measurement Error}=8.11\text{ mm} $$
Assumption: The 100 mm length is measured on a surface 30 mm above the conveyor and parallel to it, with the camera perpendicular to the conveyor.
Conclusion: Calibration using only the conveyor surface can introduce approximately 8.11% measurement error when measuring the battery's top surface. A telecentric lens is preferred for accurate dimensional inspection.

5. Photometric Correction Using Diffuse Dome Lighting

Photometric correction reduces unwanted brightness variations caused by uneven illumination, shadows and reflections. A diffuse LED dome provides light from multiple directions, reducing harsh shadows and reflections on the battery surface.
The image formation model is:
$$ f(x,y)=i(x,y)\times r(x,y) $$
Where:

- $f(x,y)$ = Observed image.
- $i(x,y)$ = Illumination component.

- $r(x,y)$ = Reflectance of the battery surface.

Taking logarithms:

$$\log f(x,y)=\log i(x,y)+\log r(x,y)$$

This separates illumination and reflectance into additive components.

Flat-field correction

A flat-field reference image is captured using uniform illumination. It represents the brightness variation across the camera's field of view.

The corrected image is calculated using:

$$I_{\text{corrected}}= \frac{I_{\text{original}}}{I_{\text{flat}}} \times \text{Mean}(I_{\text{flat}})$$

This reduces brightness differences caused by uneven lighting.

6. Python Implementation – Flat-Field Normalization

The following beginner-friendly OpenCV program demonstrates flat-field correction.

```
import cv2import numpy as np# Read original battery imageimage = cv2.imread("battery.jpg")# Read flat-field reference imageflat = cv2.imread("flat_reference.jpg")# Convert images into float valuesimage = image.astype(np.float32)flat = flat.astype(np.float32)# Calculate average brightness of flat imagemean_value = np.mean(flat, axis=(0, 1))# Avoid division by zeroflat[flat < 1] = 1# Apply flat-field normalizationcorrected = image / flat * mean_value# Keep pixel values within 0 to 255corrected = np.clip(corrected, 0, 255)# Convert back to 8-bit imagecorrected = corrected.astype(np.uint8)# Display imagescv2.imshow("Original Image", cv2.imread("battery.jpg"))cv2.imshow("Corrected Image", corrected)# Save corrected imagecv2.imwrite("corrected_battery.jpg", corrected)
```

Explanation:

1. The original battery image and flat-field reference are loaded.
2. Images are converted into floating-point values for accurate division.
3. The average brightness of the reference image is calculated.
4. Pixel-wise normalization reduces uneven illumination.
5. The corrected image is saved and displayed.

Note: The flat-field reference must be captured using the same camera, lens, aperture, lighting and exposure settings as the inspection image.

7. Technique Recommendation

Requirement	Recommended Technique
Low-cost basic inspection	Technique A
General surface inspection	Technique B
Accurate dimensional measurement	Technique C
Uniform illumination	Technique B or C
Minimum perspective error	Technique C
Reduced shadows and reflections	Technique B or C

Final Recommendation: Technique C, using an object-space telecentric lens with a diffuse LED dome, is the best choice for precision battery inspection. It minimizes perspective-related measurement errors and provides consistent illumination. Flat-field normalization and camera calibration can further improve image quality and measurement reliability.

8. Conclusion

Image formation depends on lens geometry, camera distance and illumination conditions. The calculated focal length is 8 mm for the assumed 1/2-inch sensor. However, measuring a battery surface 30 mm above the calibrated conveyor can introduce approximately 8.11% perspective error with a standard lens. Telecentric optics reduce this error, while diffuse dome lighting and flat-field correction improve brightness uniformity. These techniques help achieve reliable battery defect detection and dimensional measurement

TASK 9 – Systems Integration: Balanced Front-End Optimization Challenge [6 Marks]

Synthesize individual component choices from Tasks 1–8 into a complete operational front-end system package that fits within the ■1,25,000 budget framework. Compare three complete system packages:

- **Technique A (Budget Setup):** Rolling-shutter USB camera (1.3 MP), ambient factory line illumination, 8-bit lossy JPEG image compression archiving, running a low+mid-level software stack.
- **Technique B (Balanced Setup):** Global-shutter GigE camera (2 MP), custom LED dome light with synchronized strobe controller, calibrated standard lens, 8-bit lossless PNG image archiving, running a three-level software stack.
- **Technique C (Premium Setup):** High-speed 4k Line-scan camera, high-precision telecentric lens system, uniform line-light illuminator, 12-bit raw image archiving, running a three-level software stack.

Requirements: Create an end-to-end hardware-software pipeline block diagram showing the lifetime of a data packet from incident photons to the pneumatic rejection signal. Allocate a strict time budget (in milliseconds) to every stage to prove compliance with the 400 ms processing deadline. Construct a structured Engineering Decision Matrix comparing performance variables under extreme line conditions (e.g., passing storm clouds, fast-moving vibration, specular glare points). Include cost calculations using current industrial component price references.

1. Introduction

The aim is to combine the camera, lens, illumination, image-processing software, storage and pneumatic rejection system into one complete battery-inspection setup within the ₹1,25,000 budget.

The system must:

- Capture battery images reliably under changing factory conditions.
- Detect defects using low-, mid- and high-level image processing.
- Complete inspection within the 400 ms processing deadline.
- Send a rejection signal to the pneumatic mechanism when a defective battery is detected.
- Maintain a balance between accuracy, speed, reliability and cost.

2. Comparison of Three Complete System Packages

Feature	Technique A – Budget	Technique B – Balanced	Technique C – Premium
Camera	1.3 MP rolling shutter USB	2 MP global shutter GigE	4K line-scan
Lens	Standard lens	Calibrated standard lens	Telecentric lens
Lighting	Ambient factory light	Diffuse LED dome + synchronized strobe	Uniform line light
Image format	8-bit lossy JPEG	8-bit lossless PNG	12-bit RAW

Processing	Low + mid-level	Low + mid + high-level	Low + mid + high-level
Defect detection	Basic	Advanced	Advanced
Motion handling	Poor	Good	Excellent when correctly synchronized
Lighting variation	Highly affected	Reduced by controlled lighting	Reduced by controlled lighting
Cost	Low	Medium	Very high
Overall suitability	Limited	Best balance	High-precision applications

Observation: Technique A is inexpensive but sensitive to shadows and motion distortion. Technique C provides high-end performance but may exceed the available budget. Technique B offers a practical combination of image quality, processing capability and industrial reliability.

3. End-to-End Hardware–Software Pipeline

The recommended system is Technique B – Balanced Setup. It combines a global-shutter camera, synchronized LED lighting and three-level image processing.

1. Incident photons

Light reflected from the battery surface

2. Diffuse LED dome + strobe

Uniform illumination and motion-freezing exposure

3. Global-shutter GigE camera

Captures 2 MP image and converts light into digital pixels

4. Image acquisition and calibration

Image transfer, distortion correction and flat-field correction

5. Low-level processing

Noise removal, filtering and thresholding

6. Mid-level processing

Edge detection, contours and defect measurements

7. High-level processing

SVM / machine-learning classification: OK or DEFECT

8. Decision and PLC output

Generate accept/reject signal and track battery position

9. Pneumatic rejection mechanism

Solenoid valve activates cylinder to remove defective battery

Data packet lifetime: The captured image travels through acquisition, correction, defect analysis, classification and PLC communication. The PLC then activates the pneumatic rejection mechanism at the correct conveyor position.

4. Processing Time Budget – 400 ms Deadline

The complete inspection and rejection signal must be generated within 400 milliseconds.

Processing stage	**Allocated time**
LED strobe and image exposure	5 ms
Camera image transfer	20 ms
Image calibration and correction	15 ms
Low-level processing	20 ms
Mid-level processing	30 ms
High-level classification	40 ms
Accept/reject decision	5 ms
PLC communication and tracking	10 ms
Output and pneumatic response allowance	35 ms
Safety and timing margin	20 ms
Total	200 ms

$$ T_{\text{total}}=5+20+15+20+30+40+5+10+35+20 $$

$$ \boxed{T_{\text{total}}=200\text{ ms}} $$

200 ms – Within 400 ms deadline

Remaining time:

$$ 400-200=200\text{ ms} $$

The system therefore has a 200 ms timing margin under the assumed operating conditions.

Important: The 35 ms pneumatic allowance is a design target, not a guaranteed cylinder response time. It must be verified using the selected cylinder, valve, air pressure and mechanical arrangement. Conveyor encoder tracking is required so the rejection signal reaches the correct battery.

At 15 batteries per second:

$$ \text{Time between batteries}=\frac{1000}{15}=66.67\text{ ms} $$

The 200 ms latency does not mean the system can inspect only five batteries per second. The pipeline must process images concurrently and sustain at least 15 inspections per second.

5. Engineering Decision Matrix

The three packages are evaluated under difficult factory conditions. Scores range from 1 to 5, where 1 = poor and 5 = excellent.

Performance variable	Technique A	Technique B	Technique C
Passing storm clouds / changing light	1	5	5
Fast-moving vibration	2	4	5
Specular glare points	1	4	4
Motion distortion control	2	5	5
Dimensional accuracy	2	4	5
Defect classification	2	4	5
Processing efficiency	4	4	3
Maintenance simplicity	4	4	2
Budget suitability	5	5	1
Overall suitability	2	5	3

Explanation of the results

1. Passing storm clouds

- Technique A uses ambient light, so sudden changes in sunlight can alter image brightness.
- Technique B uses a diffuse LED dome and synchronized strobe, reducing dependence on factory lighting.
- Technique C also uses controlled line illumination, providing consistent lighting.

2. Fast-moving vibration

- Technique A uses a rolling shutter, which can create image distortion when the battery or camera moves.
- Technique B uses a global shutter, capturing the entire image simultaneously.
- Technique C can provide excellent continuous inspection when the line-scan camera is properly synchronized with a conveyor encoder.

3. Specular glare points

- Technique A is affected by reflections from shiny battery surfaces.
- Technique B reduces harsh reflections using diffuse dome illumination.
- Technique C uses uniform line illumination, but reflective surfaces may still require suitable lighting angles or polarization.

Decision: Technique B is the most balanced option because it offers reliable image capture, controlled lighting, accurate defect analysis and reasonable cost.

6. Cost Estimation and Budget Allocation

The following estimates use available Indian industrial component listings as price references and reasonable engineering allowances for components that require supplier quotations.

Technique B – Recommended Bill of Materials

Component	**Estimated cost**
2 MP global-shutter GigE camera	₹22,200
Standard calibrated industrial lens	₹8,000
Diffuse LED dome light	₹18,000
LED strobe controller	₹9,000
Processing computer	₹35,000
PLC / digital I/O module	₹8,000

Photoelectric trigger sensor	₹2,500
Pneumatic cylinder and solenoid valve	₹7,500
GigE and I/O cables	₹3,000
Mounting frame, enclosure and power supply	₹5,000
SSD storage	₹4,000
Total estimated cost	₹1,22,200

$$\text{Remaining Budget}=1,25,000-1,22,200$$

$$\boxed{\text{Remaining Budget}=₹2,800}$$

Within ₹1,25,000 budget

Price references

- A 2 MP, 90 FPS, global-shutter GigE camera is listed at ₹22,200 by Areascan Imaging.

Areascan imaging

- Industrial machine-vision lighting prices vary by design and specification. Standard LED lights are listed in an approximate ₹4,000–₹22,000 range.

Prices, Size, Forecast, and Companies

- A 2 MP GigE industrial camera is also listed by JB Microscopes at ₹28,000, providing another market reference.

Retailer from New Delhi

- Industrial 4K line-scan cameras are listed at around ₹1,50,000 for some models, while higher-performance models can cost substantially more.

Tradeindia+1

Note: The ₹8,000 lens, ₹9,000 strobe controller and other non-quoted amounts are planning allowances, not confirmed product quotations. Final procurement must verify GST, shipping, compatibility and supplier availability. The estimate assumes a basic processing PC and an existing conveyor and compressed-air supply. OpenCV and scikit-learn are open-source, so no software licence cost is included.

Estimated package comparison

Package	Estimated cost	Budget status

Technique A – Budget	₹65,000–₹75,000	Within budget
Technique B – Balanced	₹1,22,200	Within budget
Technique C – Premium	₹3,00,000 or more	Exceeds budget

Technique C is expensive because high-speed line-scan cameras and precision telecentric lenses require specialized hardware. For example, a listed 4K line-scan camera model is priced above ₹15 lakh.

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7. Final System Selection

Selected: Technique B

Technique B is selected because:

1. Global shutter reduces motion distortion.
2. Diffuse LED dome lighting minimizes shadows and glare.
3. Checkerboard calibration improves geometric measurement accuracy.
4. PNG archiving preserves image information without JPEG compression losses.
5. Three-level processing provides more reliable defect classification.
6. The estimated ₹1,22,200 cost fits within the ₹1,25,000 budget.
7. The proposed 200 ms processing and rejection-signal budget is below the 400 ms deadline.

8. Conclusion

The integrated battery inspection system combines camera hardware, controlled illumination, image processing, machine-learning classification and pneumatic rejection. Among the three packages, Technique A is economical but less reliable, while Technique C provides premium performance at a much higher cost. Technique B offers the best balance of accuracy, speed, reliability and affordability. Therefore, it is the recommended solution for industrial battery inspection within the given budget.

TASK 10 – Adaptive Engineering: High-Speed Plant Scale-Up Challenge [8 Marks]

Three months after deployment, the facility scales up. A new contract with an electric vehicle manufacturer introduces an immediate system redesign requirement:

1. The conveyor speed **doubles to 2.4 m/s** (throughput accelerates to **30 cells per second**).
2. The sorting decision window is cut in half; the system must commit an execution signal within **200 milliseconds** before cells pass the rejection gate.

3. The client requires detection of internal cell layer short-circuits. These are invisible under visible light but manifest assubtle localized thermal spots (0.5°C variations) only detectable via the Mid-Wave Infrared (MWIR) thermal spectrum.
4. The compute infrastructure remains restricted to the same 4-core edge PC without an external GPU.

Evaluate three adaptation response strategies:

- **Strategy A (Retain + Region-of-Interest Optimization):** Keep the existing hardware configuration. Shorten exposure times, down-sample image boundaries, and bound processing to highly restricted spatial Regions of Interest (ROI).
- **Strategy B (Modify Sensing and Shutter Array):** Upgrade to a higher frame-rate ultra-short global strobe area-scan camera system, implement an enclosed ambient-blocking light tunnel, and transition code arrays to INT8 format.
- **Strategy C (Redesign Architecture):** Fully transition to a multispectral/dual-stream vision system integrating a high-speed line-scan visible sensor with a specialized synchronized uncooled microbolometer MWIR thermal sensor housed within a regulated enclosure.

Requirements: Recalculate all fundamental spatial resolutions (mm/pixel), maximum allowed exposure window to prevent blur under 2.4 m/s speeds, and system data generation rates. Apply engineering logic to defend a clean **RETAIN / MODIFY / REDESIGN** architecture pivot recommendation. Evaluate the trade-offs regarding computational execution cycles, detection performance, and budget constraints. Critique how utilizing few-shot learning transfers via a pretrained Vision Foundation Model could expedite classifying new thermal defect shapes with minimal training files, while quantifying structural operational risks like domain shift, traceability failures, and model inference latency constraints on the edge PC.

1. Introduction

After three months of deployment, the battery inspection system faces three major changes:

- Conveyor speed increases from 1.2 m/s to 2.4 m/s.
- Production increases from 15 cells/s to 30 cells/s.
- The rejection decision must be generated within 200 ms.
- The system must detect internal short-circuits through very small thermal variations of 0.5°C.
- The existing computing infrastructure remains a 4-core edge PC without an external GPU.

The original visible-light inspection system cannot detect internal thermal defects. Therefore, the system must be evaluated for higher speed, lower motion blur, increased data throughput and thermal imaging capability.

2. Recalculation of Spatial Resolution

Spatial resolution indicates how much real-world distance is represented by one image pixel.

Assume the horizontal field of view remains 320 mm.

$$\text{Spatial Resolution}= \frac{\text{Horizontal FOV}}{\text{Horizontal Pixels}}$$

Using the camera resolutions considered in Task 6:

Camera resolution	Horizontal pixels	Spatial resolution
VGA	640	0.500 mm/pixel
Full HD	1920	0.167 mm/pixel
4K line-scan	4096	0.0781 mm/pixel

Example: Full HD camera

$$\frac{320}{1920}=0.1667\text{ mm/pixel}$$

This means one pixel represents approximately 0.167 mm of the battery surface.

Observation:

- VGA has low spatial resolution and may miss small defects.
- Full HD provides better detail for ordinary surface inspection.
- 4K line-scan provides finer spatial detail and is more suitable for continuous high-speed inspection.

Assumption: The calculations use a 320 mm horizontal inspection width and 4096 horizontal samples for the line-scan camera. Actual resolution depends on lens magnification, sensor configuration and calibrated field of view.

3. Maximum Exposure Time to Prevent Motion Blur

At higher conveyor speeds, the battery moves while the camera captures an image. This creates motion blur.

The motion blur equation is:

$$B=v\times t$$

Where:

- B = motion blur in mm
- v = conveyor speed in mm/s
- t = exposure time in seconds

Given:

$$v=2.4\text{ m/s}=2400\text{ mm/s}$$

For a maximum blur of one pixel:

$$t_{\max}=\frac{\text{Pixel Size}}{\text{Conveyor Speed}}$$

Exposure-time calculations

VGA:

$$t=\frac{0.5}{2400}=0.0002083s$$

$$\boxed{t=208.3\ \mu s}$$

Full HD:

$$t=\frac{0.1667}{2400}=0.00006945s$$

$$\boxed{t=69.4\ \mu s}$$

4K:

$$t=\frac{0.078125}{2400}=0.00003255s$$

$$\boxed{t=32.6\ \mu s}$$

Resolution	Pixel size	Maximum exposure for 1-pixel blur
VGA	0.500 mm	208.3 µs
Full HD	0.167 mm	69.4 µs
4K	0.0781 mm	32.6 µs

Recommended: Global shutter + short LED strobe

Engineering observation: A one-pixel blur is the theoretical upper limit used here. For more accurate defect measurement, a half-pixel blur target is preferable. This would reduce the exposure limits to approximately 104.2 µs, 34.7 µs and 16.3 µs respectively.

A short synchronized LED strobe can freeze the moving battery without requiring a long exposure. The strobe must provide enough illumination for a clear image at the selected exposure.

4. System Data Generation Rate

The production rate has increased to 30 cells per second.

Assume one image is captured for every cell and images are stored as uncompressed RGB24 (3 bytes per pixel).

$$\text{Data Rate}= \text{Image Pixels}\times3\times30$$

Data-rate calculations

VGA (640 × 480):

$$640\times480\times3\times30 =27,648,000\text{ bytes/s}$$

$$=27.65\text{ MB/s}$$

Full HD (1920 × 1080):

$$1920\times1080\times3\times30 =186,624,000\text{ bytes/s}$$

$$=186.62\text{ MB/s}$$

4K (4096 × 2160):

$$ 4096\times2160\times3\times30 =796,262,400\text{ bytes/s} $$
$$ =796.26\text{ MB/s} $$

Resolution	Data per image (RGB24)	Data rate at 30 FPS
VGA	0.922 MB	27.65 MB/s
Full HD	6.22 MB	186.62 MB/s
4K	26.54 MB	796.26 MB/s

Note: The 4K calculation assumes 4096 × 2160 pixels. A 4096 × 3000 sensor would generate approximately 1,105.92 MB/s at 30 FPS.

Effect of image format

- RGB24 requires 3 bytes per pixel.
- Mono8 requires 1 byte per pixel.
- 12-bit packed monochrome requires approximately 1.5 bytes per pixel.

For example, Full HD Mono8 at 30 FPS:

$$ 1920\times1080\times30 =62.21\text{ MB/s} $$

Using monochrome images where colour is not needed can reduce data transfer and processing requirements.

Important: These are raw acquisition rates, not necessarily the final stored-data rates. Compression, cropping and selective defect-image archiving can reduce storage requirements, but should not compromise the detection of subtle defects.

5. Evaluation of Three Adaptation Strategies

Strategy A – Retain + Region-of-Interest Optimization

RETAIN

This strategy keeps the existing camera and computing hardware while optimizing the software.

Methods:

- Reduce exposure time to minimize motion blur.
- Process only important Regions of Interest (ROI).
- Down-sample unnecessary image boundaries.
- Use smaller image arrays.
- Optimize OpenCV operations and avoid unnecessary memory copying.
- Use parallel processing where practical.

Advantages:

- Lowest additional cost.
- Existing hardware and software can be reused.
- Reduced processing time and memory usage.
- Faster implementation.

Limitations:

- Existing visible-light camera cannot detect internal thermal short-circuits.
- Ambient illumination remains sensitive to changing factory light.
- ROI optimization cannot compensate for missing thermal information.
- A 4-core CPU may struggle with 30-cell-per-second image processing.

Evaluation: Strategy A is useful for improving existing visible-light inspection, but it cannot satisfy the new thermal-defect requirement. Therefore, it is not a complete solution.

Strategy B – Modify Sensing and Shutter Array

MODIFY

This strategy upgrades the image acquisition hardware while retaining the existing edge-computing architecture.

Methods:

- Use a high-frame-rate global-shutter camera.

- Install an ultra-short synchronized LED strobe.
- Enclose the inspection area in an ambient-blocking light tunnel.
- Convert suitable neural-network operations to INT8.
- Use ROI-based image processing.
- Optimize data transfer and memory allocation.

Advantages:

- Reduces motion distortion.
- Reduces the effect of changing external light.
- Improves visible defect detection.
- INT8 inference can reduce memory use and improve CPU execution speed when supported by optimized libraries.
- Existing software architecture can be retained with modifications.

Limitations:

- Visible-light cameras cannot directly detect internal thermal spots.
- INT8 conversion may reduce classification accuracy if not calibrated and tested.
- Higher frame rates increase data-transfer and memory-bandwidth demands.
- The existing four-core PC remains a computational limitation.

Evaluation: Strategy B is a good improvement for visible-light inspection but still cannot meet the client's requirement for detecting internal thermal short-circuits.

Strategy C – Redesign Architecture

REDESIGN

This strategy replaces the single-stream visible inspection system with a synchronized multispectral inspection architecture.

It combines:

1. High-speed visible line-scan camera.
2. Specialized thermal camera.
3. Encoder-based synchronization.
4. Regulated thermal inspection enclosure.
5. Lightweight dual-stream processing software.

Moving battery cell

Conveyor encoder provides position and timing

Visible stream

High-speed line-scan camera

Surface scratches and cracks

Thermal stream

MWIR thermal camera

Localized thermal anomalies

Synchronization and calibration

Align images using encoder position and timestamps

Dual-stream feature extraction

Visible defect features + thermal anomaly features

Feature fusion and classification

Combine both streams and classify the cell

PLC → Pneumatic rejection

Reject defective cell at the correct conveyor position

Advantages:

- Detects both visible surface defects and internal thermal anomalies.

- Supports continuous high-speed inspection.
- Provides better defect coverage.
- Synchronization allows visible and thermal information to be associated with the same battery.

Limitations:

- Higher camera and integration cost.
- More complex calibration and synchronization.
- Thermal imaging requires controlled operating conditions.
- Four-core CPU processing must be carefully optimized.
- Thermal sensors require validation for the specified 0.5°C temperature variation.

Important technical correction: Uncooled microbolometer cameras are generally long-wave infrared (LWIR) devices, commonly operating around 8–14 µm. MWIR imaging commonly uses specialized photon detectors and is often more expensive, frequently requiring detector cooling. Therefore, the proposed MWIR camera must be selected according to the actual spectral requirement; an ordinary uncooled microbolometer should not automatically be treated as an MWIR sensor.

6. Processing Deadline and Computational Execution

The new requirement is to generate the rejection execution signal within 200 ms.

At a conveyor speed of 2.4 m/s:

$$d=v\times t$$

$$d=2.4\times0.2$$

$$\boxed{d=0.48\text{ m}=480\text{ mm}}$$

This means the battery travels 480 mm in 200 ms.

Therefore, the rejection gate must be positioned sufficiently far downstream, allowing time for image processing, PLC communication and actuator response.

Proposed processing-time allocation

Processing stage	Time budget
Image acquisition	10 ms
Image transfer	15 ms
Calibration and preprocessing	10 ms
Visible-stream feature extraction	20 ms
Thermal-stream feature extraction	25 ms
Feature fusion and classification	35 ms
Decision generation	5 ms
PLC communication and tracking	10 ms
Pneumatic response allowance	35 ms
Safety margin	25 ms
Total	190 ms

$$T_{\text{total}}=190\text{ ms}$$

$$200-190=\boxed{10\text{ ms remaining}}$$

This is a proposed engineering budget, not a measured benchmark. Actual camera transfer, thermal processing and pneumatic response times must be tested on the selected hardware.

Four-core edge PC optimization

Since an external GPU is not available, the system should:

- Use INT8 quantization for supported neural-network models.
- Apply ROI processing instead of processing every pixel.

- Use OpenCV optimized CPU operations.
- Separate acquisition, preprocessing, inference and PLC communication into controlled processing threads.
- Avoid unnecessary image copies.
- Use lightweight models instead of large deep-learning architectures.
- Reserve sufficient CPU capacity for real-time control and communication.

Throughput requirement:

$$\text{Time between cells}=\frac{1000}{30}$$

$$\boxed{33.33\text{ ms per cell}}$$

The system must accept a new cell every 33.33 ms, even though an individual cell's decision may take up to 200 ms. Therefore, processing must be pipelined, and multiple cells may be in different processing stages simultaneously.

7. Few-Shot Learning Using a Pretrained Vision Foundation Model

Few-shot learning is a machine-learning technique that enables a model to recognize a new defect category using only a small number of labelled training examples.

In this project, internal short-circuit thermal defects may have very few available training images because such failures are uncommon.

A pretrained Vision Foundation Model can help extract useful image features from a small dataset.

Working process

1. Pretrained model: Start with a model that has already learned general visual patterns.
2. Thermal image collection: Capture thermal images of normal cells and cells with known thermal defects.
3. Feature extraction: Convert thermal images into feature vectors.
4. Few-shot training: Provide a small number of labelled defect examples.
5. Classification: Compare new thermal images with learned defect patterns.
6. Validation: Test the model using unseen thermal images.

Pretrained Vision Foundation Model

Small labelled thermal dataset

Normal cells + thermal defect examples

Feature extraction and few-shot adaptation

Thermal defect classifier

Normal / Possible short-circuit

Advantages

- Requires fewer labelled examples than training a model from scratch.
- Reduces the time needed to introduce new defect categories.
- Can reuse learned image features.
- Supports faster development when thermal defect examples are limited.

Limitations and operational risks

Risk	Explanation	Mitigation
Domain shift	A model pretrained on ordinary RGB images may not understand MWIR thermal patterns correctly.	Use thermal-pretrained models or adapt and validate the model using real MWIR images.
Small dataset	A few examples may not represent all possible short-circuit shapes and temperatures.	Collect examples across operating conditions and use independent validation data.
False negatives	A subtle 0.5°C anomaly may be missed.	Validate sensitivity using calibrated thermal references and known defect samples.
Traceability failure	Model updates or incorrect image-to-cell association may produce unreliable inspection records.	Store image ID, cell ID, timestamps, model version and decision logs.
Inference latency	Large foundation models may exceed the edge PC's available processing time.	Use a compact feature extractor, quantization and ROI-based inference.

Temperature variation	Ambient temperature and battery operating conditions may affect thermal readings.	Use a regulated enclosure and temperature calibration.

Quantifying inference constraints

At 30 cells per second, the system receives one cell every 33.33 ms.

For a single sequential inference worker, the average processing time must be below 33.33 ms per cell to keep up without an accumulating queue.

A 4-core CPU may use multiple workers, but the achievable throughput depends on memory bandwidth, model size and thread contention. Parallelism cannot be assumed to provide a fourfold speed increase.

A suitable engineering target is:

- Lightweight thermal feature extraction: 10–15 ms.
- Thermal classification: 10–15 ms.
- Visible and thermal feature fusion: 5 ms.

These are target allocations, not guaranteed model benchmarks. The complete processing pipeline must be profiled on the actual edge PC.

Recommendation: Use a compact, thermal-adapted model rather than directly deploying a large pretrained foundation model on the four-core PC. Few-shot learning can reduce training effort, but it does not eliminate the need for thermal-specific validation.

8. RETAIN / MODIFY / REDESIGN Decision

Strategy	Main benefit	Main limitation	Decision
A – Retain	Low cost and faster software processing	Cannot detect internal thermal defects	RETAIN existing visible inspection only
B – Modify	Better speed, lighting and motion handling	Still cannot detect internal thermal defects	MODIFY visible sensing
C – Redesign	Combines visible and thermal defect detection	High cost and integration complexity	REDESIGN complete inspection architecture

Final architecture pivot

Existing visible inspection

RETAIN

Keep useful OpenCV algorithms, calibration methods, defect rules and existing inspection records.

Camera and illumination

MODIFY

Upgrade to high-speed global-shutter or line-scan sensing, synchronized illumination and encoder-based triggering.

Thermal detection architecture

REDESIGN

Introduce a synchronized, appropriately specified thermal camera and a separate thermal anomaly detection pipeline.

9. Budget and Engineering Trade-Offs

Strategy	Relative additional cost	Performance
A – Retain	Low	Improved visible processing only
B – Modify	Medium	Better high-speed visible inspection
C – Redesign	High	Visible and thermal inspection

The exact cost of Strategy C depends heavily on the selected MWIR detector, optics, frame rate, thermal sensitivity, enclosure and integration requirements. A genuine MWIR system may require a substantial investment and could exceed the original ₹1,25,000 budget.

The four-core edge PC should be retained initially, but its ability to sustain both image streams must be established through a prototype benchmark. If it cannot meet the 30-cell/s throughput and 200 ms decision requirement, the compute

architecture will also need upgrading or the acquisition and preprocessing stages will need dedicated hardware acceleration.

10. Conclusion

The scale-up from 15 to 30 cells per second requires faster image acquisition, shorter exposure times, optimized data transfer and efficient processing. At 2.4 m/s, the maximum one-pixel-blur exposure is approximately 69.4 µs for Full HD and 32.6 µs for the assumed 4K resolution. The system must also maintain a 200 ms decision deadline while processing a new cell every 33.33 ms.

Strategy A can improve the existing software, and Strategy B can improve visible-light inspection, but neither can detect internal thermal short-circuits. Therefore, Strategy C is the necessary long-term architecture redesign, combining synchronized visible and thermal imaging. Few-shot learning can accelerate the introduction of new thermal defect categories, but thermal domain shift, traceability and CPU inference latency must be carefully controlled. The final system should retain useful existing software, modify visible sensing and redesign the inspection architecture to include thermal detection.

Evaluation Grading Metric Rubric

Task Number	Focus Core Module Domain	Max Marks
Task 1	Overview and Features of Computer Vision	4
Task 2	Levels of Computer Vision and Image Processing	5
Task 3	Digital Image Fundamentals I: Representations	4
Task 4	Digital Image Fundamentals II: Pixel Connectivity	4
Task 5	Image Sensing and Acquisition Systems	5
Task 6	Sampling and Spatial Limits	5
Task 7	Quantization and Intensity Limits	4
Task 8	Image Formation Models (Geometric/Photometric)	5
Task 9	Applications and Integrated Edge Systems	6
Task 10	Re-apply and Adapt (Engineering Scaling Challenge)	8
TOTAL	**Comprehensive Evaluation Score Matrix**	**50**